



Module Title
Second Year Project

Assessment Weightage & Type
Second Year Project
2025-26 Autumn

Members	University ID
Samir Nepal	24046042
Ganga Bajgain	24045880
Manish Bhattarai	24045919
Sushant Somphoru	24046121

Submitted To: Kritika Chaudhary

Project Title: SajiloDocs

Assignment Due Date: Friday, March 27, 2026

Assignment Submission Date: Friday, March 27, 2026

Word Count: 10799

I confirm that I understand my coursework needs to be submitted online via mySecondTeacher under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

Acknowledgement

We would like to express our sincere gratitude to our project supervisor, Miss. Kritika Chaudhary, for their constant guidance, encouragement, and valuable suggestions throughout the entire process of this project. Their knowledge, patience, and support played a vital role in helping us shape our understanding and successfully complete this work.

We are also deeply thankful to Itahari International College and our department for providing a supportive academic environment, along with the necessary resources and facilities to carry out this project. The knowledge and skills we gained during our studies greatly contributed to the successful completion of this work.

We would like to extend our appreciation to our team members for their dedication, cooperation, and teamwork. Their ideas, encouragement, and collaboration added great value and meaning to this project.

We are equally grateful to everyone who directly or indirectly contributed to this project through their suggestions, feedback, and support.

Finally, we would like to thank our family and friends for their constant encouragement, understanding, and belief in us. Their support motivated us to overcome challenges and successfully complete this project.

Abstract

It is a project, which is named SajiloDocs and developed to solve the increasing problem of losing, spoiling, or poorly maintaining important physical documents like academic certificates, identification cards, property papers, and other legal documents. In traditional systems, users are inclined to use manual ways of storing data and this is time-consuming, prone to mistakes and is poorly secured. There are also other general cloud storage platforms that are not very user-specific and may not address the needs of individual users with specific awareness, like the localized documents such as Nepali language documents, among others.

The proposed system can address such problems by proposing a secure and friendly web based digital locker that would help users to upload, store, organize and maintain their documents effectively. The system also offers categorization and allows the users to have a structured storage where they are able to access their files whenever they require them. It also comes with secure authentication system to prevent sensitive user information and prevention of unauthorized access. A key feature of this system is the integration of Optical Character Recognition (OCR) technology, which enables the platform to extract text from scanned or image-based documents. Moreover, the system can automatically transform extracted Nepali text into English or many languages which allows to make documents more understandable and easier to access. The option of reading the translated contents through a well-formatted PDF format and downloading them is also available to the users.

The system also facilitates controlled file sharing, which allows users to create secure links and the permission level can be assigned, including view-only or edit access. Another important functionality is the expiry alert system, which reminds them when their important documents are expiring, enabling them to make the necessary action in time.

The project is developed using modern web technologies, such as React in the user interface, Django in the backend processing, and MySQL in database management. OCR tools like Tesseract OCR, third-party translation APIs are also included to increase system choices. The development is done in a systematic approach so as to be reliable, performing, and scalable.

The end result of the project is a practical, efficient and secure digital document management system that is user friendly, specifically in Nepal. It will minimize human resource, enhance the accessibility of documents, secure data, and offer an easy user experience. In general,

27 March 2026

SajiloDocs is an extensive platform to keep the most valuable papers in a safe, well organized, and smart way.

1 Table of Contents

1. Introduction.....	1
1.1 Introduction to Project	2
1.2 Identification of Client.....	2
1.3 Problem Scenario	2
2 Literature Review/Background Study.....	3
2.1 Timeline of Report/ Project Gantt Chart.....	3
2.2 Project as a Solution.....	6
2.3 Aims And Objectives	7
2.4 Out Comes and Deliverables	8
2.5 Similar Projects.....	9
2.5.1 Project 1: Microsoft OneDrive.....	9
2.5.2 Project 2: Google Drive	9
2.5.3 Project 3: Dropbox	10
2.5.4 Similar System Comparison	11
2.5.5 Critical Evaluation	12
2.6 Review Summary.....	13
2.7 Project Risks, Threats, and Contingency Plan	13
2.7.1 Risk and Threats	14
2.7.2 Contingency Plan.....	14
2.8 Resource Requirements	16
2.8.1 Hardware Requirements.....	16
2.8.2 Software Requirements	17
3 Methodology	18
3.1 Considered Methodology.....	18
3.2 Chossen Methodology	19
4 System Design	20

4.1	System Architecture.....	21
4.2	Entity Relationship Diagram(ERD).....	22
4.3	Use Case Diagram.....	24
4.4	Flow Chart	26
5	Software Requirements Specification (SRS).....	Error! Bookmark not defined.
6	Testing.....	28
6.1	Test Case 1	29
6.2	Test Case 2	30
6.3	Test Case 3	30
6.4	Test Case 4.....	31
6.5	Test Case 5.....	31
6.6	Test Case 6.....	32
7	Group Contribution Table.....	32
8	Conclusion	38
8.1	Legal Issues.....	39
8.2	Social Issues.....	39
8.3	Ethical Issues	40
8.4	Future Work.....	41
9	References.....	42
10	Appendix.....	Error! Bookmark not defined.
11	Developed Project.....	44
11.1	Screenshots	44
11.2	UI Designs	49

Table of figures

Figure 1: Gantt Chart	5
Figure 2: System Architecture of Sajilodocs	21
Figure 3: Login Page.....	44
Figure 4: Sign Up Page	44
Figure 5: Dashboard Page.....	45
Figure 6: Create folder.....	45
Figure 7: Upload File	46
Figure 8: Notifications Page	46
Figure 9: Docx Edit Page.....	47
Figure 10: Image View	47
Figure 11: Edit Profile Page.....	48
Figure 12: Dashboard UI	49
Figure 13: Document View UI	50
Figure 14: Home Page UI	50
Figure 15: Login UI.....	51
Figure 16: Notifications UI.....	51
Figure 17: Edit Profile UI	52
Figure 18: Give Access UI.....	53

Table of tables

Table 1 Similar System..... 11

Table 2 Contingency plan 15

Table 3 Hardware Requirements 16

Table 4 Software Requirements..... 17

Table 5: Test 1- User Registration29

Table 6: Test Case 2- User Login30

Table 7: Test Case 3-Document Upload.....30

Table 8: Test Case 4-Text OCR Extraction31

Table 9: Test Case 5-Document Translation and PDF Generation.....31

Table 10: Test Case 6-Expiry Notification System32

Table 11: Group Contribution Table.....32

1. Introduction

This chapter introduces the project SajiloDocs and the context which it has been developed. The issue of handling valuable personal and legal documents has been a regular challenge to many people in the digital age. Such documents as ID, academic certificates, and legal documents can be disorganized, dispersed over various digital devices, and are not easy to organize, extract, and protect.

The chapter justifies the background of this issue by pointing at the limitations of the traditional storage method and general cloud-based solutions. These existing approaches are often time-consuming, unsecure, and lack intelligent functions like document categorization, expiration and language translation. This turn makes users struggle to access documents fast, keep them in good condition and share in safety whenever the need arises.

The major stakeholders of the project are the general users like students, professionals, family, and the owners of small business who require a stable system to handle their valuable documents. The necessity to develop a useful digital solution is connected with the growing demand to find secure, systematized, and easily available document management systems.

Therefore, the proposed project includes a user-friendly and intelligent digital solution, that addresses these all challenges by the centralizing document storage and introducing advanced features like OCR, translation and secure sharing.

Brief overview in the other sections of the report such as literature review, methodology, system design, software requirements specification (SRS), testing, and final implementation details are also given in this chapter. All these sections combine and describe how the project is developed, evaluated, and successfully implemented.

1.1 Introduction to Project

SajiloDocs is a web-based system and developed to improve the way of store and manage and access the important private documents. Project connected to the domain of digital document management whereby secure storage, easy access and efficient information management is necessary. The proposed of this system is to replace the traditional physical document storage methods by providing the reliable and user-friendly as well as organized the digital platform. It can allow the user to upload the documents categorize them and extract the text using OCR technology and translate them into many languages. With this system ensures the better accessibility and improved the organization and enhanced security for the sensitive information.

1.2 Identification of Client

The primary customer of such a project is general public, students, professionals and owners of small businesses who need a secure option of keeping their valuable documents. These users find it difficult to store and keep physical records in a safe manner. This system is developed taking in consideration the requirements of such users like the ability to store their data securely, ability to retrieve it easily and sharing of documents. It also takes care of the user requirements of simplicity, reliability and accessibility. Hence, SajiloDocs is a convenient tool that can be personalized to address the daily document management requirements of its intended customers.

1.3 Problem Scenario

At present the most important documents like academic certificates, identification cards and legal papers are either stored manually or dispersed in general cloud storage systems. This leads to a number of issues such as the inability to organize, the possibility of losses or damage and the time-consuming retrieval. Besides, the users usually have difficulties in manipulation of the documents of other languages, particularly Nepali documents that need to be translated. Manual handling enhances possibilities of mistakes, duplication and inefficiency. These problems lead to low productivity as well as inconvenience among consumers.

This state of affairs suggests that it is necessary to have a more organized, secure and technology-oriented solution such as SajiloDocs that will be able to centralize document storage, offer intelligent capabilities such as OCR and translation and enhance the overall user's experience.

2 Literature Review/Background Study

The chapter offers theoretical and practical framework that is required in formulation of the Digital Locker System. It is devoted to the discussion of the pertinent concepts, existing systems, and previous research associated with the digital document management, data security, and cloud-based storage solutions. The aim of this section is to show the clear knowledge of the area and make certain that the given system is grounded on sound research and not personal assumptions. Digital document management systems are gaining significance as the level of digital transformation increases. Use of traditional means of storing documents in the physical form is inefficient and can easily be lost and managed. The current systems like cloud storage systems and digital vaults have solutions to store and access documents online, but most of them do not have advanced mechanisms like smart categorization, text extraction through the use of OCR, and support of the specific language.

Research studies emphasise on secure storage systems such as encryption, authentication and role-based access controls as a way of safeguarding sensitive information. There is also the use of the Optical Character Recognition (OCR) technology that has been extensively used in converting documents that have been scanned to editable and searchable text which has enhanced the accessibility and usability of documents.

Some contemporary systems also include such features as programmed reminders, sharing of documents, and real time access which make it user friendly and more convenient to work with other individuals. Nevertheless, systems in which such functions are combined in one people-friendly application remain a gap, particularly when they have support to local languages like Nepali.

This study determines the current practices, limitations, and opportunities of improvement by examining these available solutions and technologies. These lessons have informed the design and development of the proposed Digital Locker System such that the system meets real life challenges and offers a viable, secure, and intelligent system of managing documents.

2.1 Timeline of Report/ Project Gantt Chart

The Gantt Chart gives a detailed graphical account of the project schedule and timeline of the project development of SajiDocs platform. It describes all significant stages, sprints and activities along with an approximate start and finish date, indicating clearly the sequence in which each activity starts and finishes till the final release (LaPrad, 2026). The project consists of various important phases. Starts with the Initiation phase November 2025 start, with

Requirement Gathering, Product Backlog, project methodology selection, Project Charter, and UI/UX Design. This is a background stage that determines the course of the whole project.

After the beginning, the project passes to the stage of the Iteration performed using a series of agile sprints. Every sprint is about the delivery of certain functionalities:

- Sprint 1: Covers user authentication features including user registration login with OTP JWT token management and Authorization testing.
- Sprint 2: is based on basic Document CRUD functions, Document API, and file upload features.
- The Sprint 3: features the Access Control including Permissions API, Share Link Generation and Access Log Tracking.
- Sprint 4: combines AI-enhanced OCR (Optical Character Recognition) features such as the ability to include OCR engines and extract text functions.
- Sprint 5: comes with the Translation module that supplies Nepali English and many other languages translation and PDF re-creation.
- Sprint 6: comes with added functionality of Search and Filter using full-text search.
- Sprint 7: will also have Reminder and Notification systems such as in-app alerts and email.
- Sprint 8: this version is dedicated to Security improvement that includes encryption, RBAC (Role-Based Access Control) hardening, activity logging, and security testing.

During the development, frequent Sprint Review and Retro meetings will be held to assess the progress, discuss the problems, and enhance the development process. The last phase of the project will be the Release phase towards end of March 2026 that will involve final Deployment and final Documentation.

The Gantt Chart aptly illustrates that the project is developed in an iterative and incremental process based on the Agile methodology. The majority of the core features will be done by December 2025 to February 26, but the security and final releases will take place in March 2026. This schedule will make sure that there is a systematic development, testing of a good document management system and that the system will be delivered in good time. (Wadhwa, 2024)

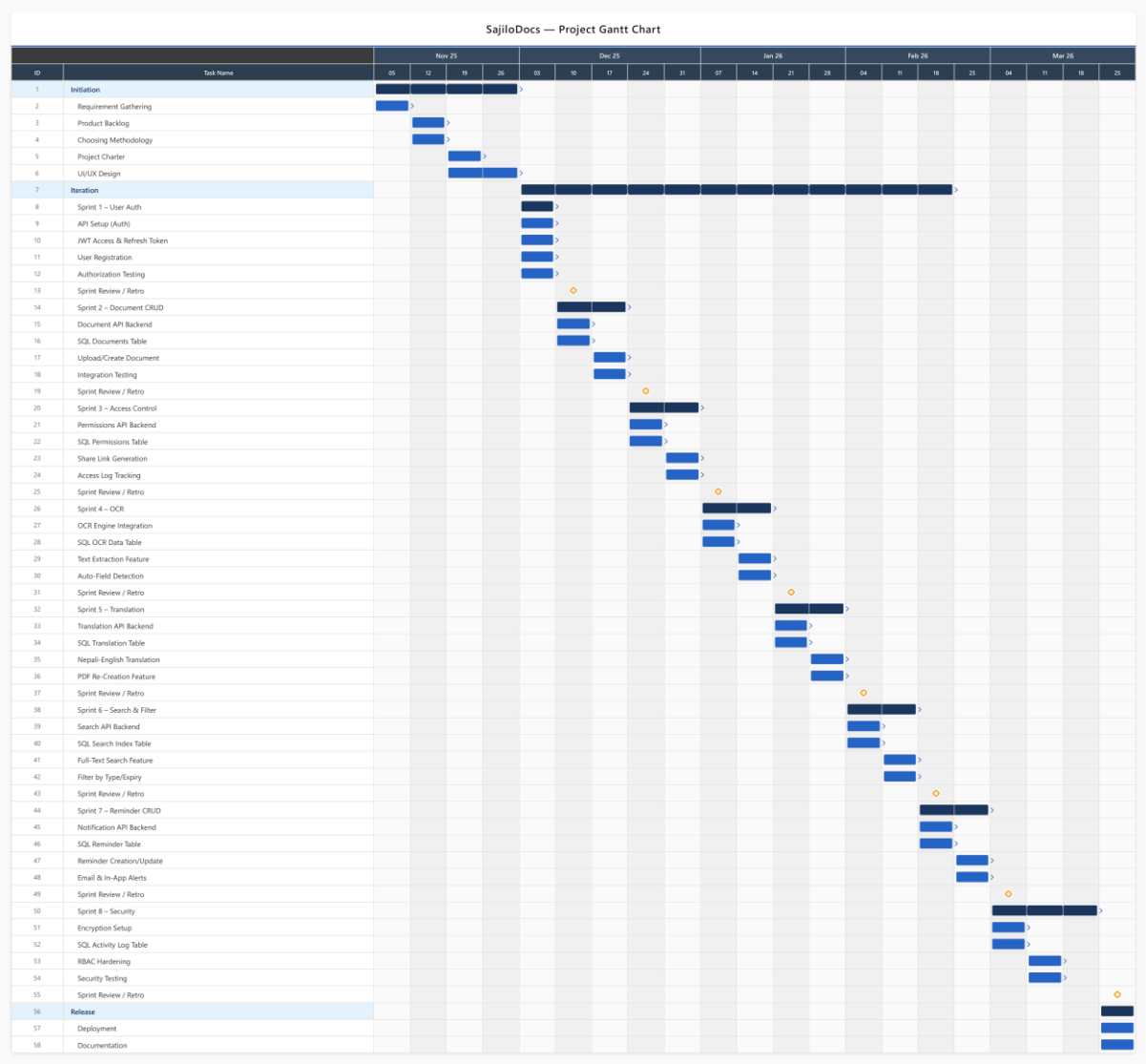


Figure 1: Gantt Chart

2.2 Project as a Solution

The proposed Digital Locker System can be used as the remedy to the issues of mismanagement of the documents, their loss, and security threats that were discussed in the previous sections. The conventional systems of managing valuable personal and legal information are usually based on the manual storage of the documents, which may result in the lack of organization, inaccessibility and possible privacy concerns. To overcome these struggles, the system provides a secure web based system which centralizes repository and administration of documents in an organized user friendly system. The system offers smart capabilities like indexing of metadata and document categorization, text extraction through OCR, instead of using the physical files or a collection of disjointed digital records.

These characteristics enable users to easily arrange, search, as well as access documents with a minimum effort. Also, the translation and document recreation functions in Nepali-to-English make sure that the users have access to the clear and standardized versions of professionally formatted documents, when required.

The system has automated expiry-date warning notifications, file-sharing with security, and access control by roles to achieve additional usability and reliability. These characteristics mitigate the possibility of missing crucial deadlines, enhance team work, and make sure that crucial data is accessible to the authorized personnel. The system reduces the human error through strong security mechanism coupled with automation, and improves the general protection of data.

Consequently, the Digital Locker System enhances a high level of accessibility, the security of documents, and the decrease of manual work, as well as the effective management of information. It is also a good easy to use and reliable solution to students, families and small organizations because it helps resolve both operational issues and user-related troubles.

2.3 Aims And Objectives

The main aim of the project is to design and develop a secure and intelligent Digital Locker System of efficient storage, management, and sharing of personal and legal documents. The system is designed to offer centralized, reliable and intuitive platform that enhance accessibility, increase security as well as minimize the hassles involved in the process of manual document handling.

To achieve this aim, the project has the following objectives:

- To examine and investigate the current issues connected with the document storage, organization and security.
- To identify and analyse the user needs to a digital document management system.
- To develop a system architecture that is well structured and user friendly to store and organize documents.
- To come up with some basic functionality including document upload, categorization, and metadata indexing.
- To add text extraction and Nepali to English translation features built upon OCR.
- To add security features like the role-based access control and secure file sharing.
- To add further features such as expiry-date reminders and recreation of the documents.
- To test the system in terms of accuracy, usability, performance, and security.

2.4 Out Comes and Deliverables

The identified project success factor is that the Digital Locker System will be developed in a successful way so that it offers a safe, structured, user-friendly system to store, administer and share personal and legal documents. The system will empower the users to upload their documents, classify, search and retrieve them easily and securely maintain the data privacy and security. The project will provide a web-based application that will have the necessary features of document upload, metadata indexing, OCR-based text extraction, Nepali-to-English translation, secure file sharing, role-based access control, and automatic expiry-date notifications. All these aspects will complement usability, accessibility and reliability of document management.

The major deliverables of this project are:

- A complete Digital Locker System (web application).
- Whole system source code.
- Implemented screens and designs of user interface.
- Document management and storage database.
- System architecture/ design drawings.
- Software Requirements Specification document (SRS).
- Final report on project development and implementation.
- Performance, usability, and security Test results are used to show that the system performs, is usable and is secure.
- End-user instructions manual.
- System demonstration presentation material.

These deliverables are the results of the project both academic and practical, such that the system is not only sound and technical but also focused on the user and solves real-life document management issues.

2.5 Similar Projects

A number of existing systems concerning the cloud-based document storage and management were analysed in order to get to know existing practices and trends in features. These platforms reveal the existing methods of storage, accessing and sharing of digital files by users. Nonetheless, an overview of each system makes it clear that there are major constraints that justify the idea of SajiloDocs being a better and more focused and feature-oriented tool to manage personal and legal documents in Nepal.

2.5.1 Project 1: Microsoft OneDrive

OneDrive is the Microsoft cloud service that connects you to all your files. It lets you store and protect your files, share them with others, and get to them from anywhere on all your devices.(Microsoft, 2026) It is closely tied to the Microsoft 365 suite to allow customers to save their documents, photos, and other files on the Web and use them on any device. OneDrive allows working on the Word, Excel, and PowerPoint files in real time on the computer using only the browser. It also provides versioning, recycle bin recovery and sharing control that enables users to identify view-only or edit permission of shared content.

It is also a big strength since it can easily be integrated with Windows operating systems and Microsoft office applications which means that it is an automatic choice amongst business and academic users who are already in the Microsoft ecosystem. OneDrive is however a general-purpose storage service that does not have document intelligence. It will not give automated expiry notice of time-based documents, no OCR facilities to work with non-English or handwritten documents, and to work with Nepali language documents. It requires users to scan the files which contain no structured metadata or categorization according to personal legal needs.

2.5.2 Project 2: Google Drive

Google Drive is a cloud-based storage service that enables users to store, access, and share files and folders online. The web service provides a centralized cloud storage hub where changes to content are automatically saved and synchronized across all connected devices. (Hanna, 2026)It allows users to upload, share and edit documents in real time. Google drive also has a simple search feature that can search a text in PDF files using the inbuilt Google OCR feature and therefore the scanned files are partially searchable.

Google drive, although featuring a wide range of features and being easy to use, is structured as an overall storage system and is not aimed at the more structured needs of personal and legal document management. Although it provides simple OCR on the English-language documents it does not provide the accurate extraction of text in the Nepali documents, as well as provision of automatic translation feature. Document expiry dates, reminders and notarized version of uploaded files have no mechanism of tracking. Access to sharing is done via simple permission links instead of a role-based access control system, which has revocable token-based access.

2.5.3 Project 3: Dropbox

Dropbox is a place where all your team's content comes together. Where you can use the tools you love. Where we help you cut through the clutter and surface what matters most.(Dropbox, 2026) Some of the features offered are Smart Sync to selectively store files locally, Dropbox Paper to collaboratively create documents, shared team folders, and file recovery features. Businesses and freelancers are the ones who are most likely to utilize Dropbox as a means of getting a reliable, prompt file synchronization platform with numerous devices.

Nevertheless, Dropbox is placed as a universal file syncing and collaboration service that lacks document intelligence and language processing. It lacks OCR capabilities, the ability to translate documents, and the possibility to extract structured metadata of uploaded documents. It does not have any inbuilt expiry tracking, or document renewal reminders. With personal data (citizenship certificate, passport, academic transcript) and individual user, there are no context-sensitive facilities of dropbox, and the user must handle the organization all by themselves. Its pricing structure also renders it inaccessible to individual users in the developing regions.

2.5.4 Similar System Comparison

The table below provides a structured comparison of the three reviewed systems against SajiloDocs across key criteria including purpose, features, language support, access control, and limitations.

Table 1 Similar System

Criteria	OneDrive	Google Drive	Dropbox	SajiloDocs
Main Purpose	General file storage & MS Office integration	Cloud storage & Google Workspace collaboration	File sync & team collaboration	Smart document locker for personal/legal docs
Key Features	File storage, Office online editing, sharing, versioning	File storage, Docs/Sheets editing, sharing, search	File sync, Paper, Smart Sync, team folders	OCR extraction, Nepali translation, expiry reminders, RBAC sharing, notarized PDF
OCR Support	Basic (Microsoft Lens integration only)	Basic (Google Drive PDF/image text search)	None	Full OCR using Tesseract, with field detection
Nepali Language Support	None	Limited (Google Translate plugin)	None	Built-in Nepali-to-English translation & recreation
Expiry Reminders	None	None	None	Automated alerts 7, 15, 30 days before expiry
Access Control	Basic link sharing with permission levels	Basic link sharing with permission levels	Basic link sharing, team folders	Role-based (View, Download, Edit) with token-based access
Document Recreation	None	None	None	Generates formatted, translated PDF

Target Users	General public, businesses, Office 365 users	General public, students, Google Workspace users	Teams, businesses, professionals	Nepali individuals, students, families, small orgs managing personal/legal documents
Limitations	No document-specific metadata, no OCR for Nepali, no expiry tracking	No expiry tracking, no Nepali OCR, no structured legal doc management	No OCR, no translation, no expiry reminders, expensive for individuals	Currently local deployment only; limited to web platform

2.5.5 Critical Evaluation

Comparison of OneDrive, Google Drive, and Dropbox indicates that they are all technically sound and are used extensively, but a similarity between these platforms is that they are all horizontal storage systems and not intelligent and context-sensitive document management systems. The reviewed sites do not have a built-in expiry warning system, Nepali OCR services, document translation, or the power to recreate a formatted legal document in another language.

Moreover, the current systems lack organized metadata management that fits personal legal records, i.e., document type, date of issuance, date of expiration, or the category of government. Their access control systems are functional, but are not fine-grained role or activity logs and are basic in nature (only link-based sharing). To Nepali users who need to handle documents like citizenship certificates, academic records and property papers, these platforms do not offer the localized and intelligent capabilities required.

SajiloDocs directly fills these gaps by providing secure cloud storage with the functionality of OCR-based text extraction, automated Nepali-to-English translation, expiry management, role management, and downloadable notarized documents. This makes SajiloDocs not a direct competitor of general-purpose storage software, but an environmentally specific and complementary product that addresses the document management requirement of Nepali people, students, and small organizations.

2.6 Review Summary

Overall, the design and development of SajiloDocs were well based on the background research and analysis of other systems in the field. The examination established that the capability of current cloud storage platforms, i.e., OneDrive, Google Drive, and Dropbox, is helpful in general-purpose applications, yet it is always deficient when it comes to serving the distinct requirements of the users who deal with personal, legal, and language-related documents.

The review revealed four crucial gaps that cannot be satisfied by any existing platform: no support of the Nepali language OCR and translation, no automated expiry notices of time-sensitive documents, no organized grouping of documents by legal metadata, and no notarized downloadable document productions. These results confirm the relevance of SajiloDocs as a purposeful solution and have a direct impact on feature set, technology selection, and design priorities of the system, which are presented in the subsequent chapters.

2.7 Project Risks, Threats, and Contingency Plan

their potential impact, and outlines a contingency plan to manage each one effectively.

With software development project, there are uncertainties which are intrinsic and can influence the timeline, quality or the end product of the project. In the case of SajiloDocs, which unites several modules that are difficult to implement, such as OCR processing, machine translation, role-based access control, and PDF generating, it is necessary to carefully determine the risks and plan beforehand. The next section will establish the key risks and threats of the project, evaluate their probable effect, and provide a plan of action to deal with each of them.

2.7.1 Risk and Threats

The significant risks and threats that were discovered in the SajiloDocs project fall under the categories of technical, operational and team-based. The delays encountered might be because of the academic load of all team members especially when they are doing assessment which could lower the time of completion of the sprint. Technical risks are the possibility of inaccuracy of Tesseract OCR processing Nepali handwritten or printed text, and reliance on third-party translation APIs that can be offline or have bandwidth restrictions. The unauthorized access to stored documents is among data security risks that can weaken the trust of the user and share sensitive personal information. The risk associated with a team is that one of the members may be unavailable temporarily owing to illness or other personal commitments and this would lead to the failure of delivery of the sprints in case the distribution of responsibility is not well spelled out.

2.7.2 Contingency Plan

In order to mitigate the effect of the identified risks, the following contingency plans are included in the project plan. The Scrum methodology is flexible by definition as the structure of the sprint system enables the team to prioritize tasks again in case of delays. Stand up meetings are also taken weekly so that problems are detected and addressed at an early stage. The periodic commits to Git and PostgreSQL database backups safeguard against information loss. This has been discovered to have a fallback translation API in case the primary service is not available. JWT-based authentication, role-based access control, encrypted file storage, and activities logging help in minimizing security risks. Each module has been briefed to all the team members to make them cross cover in the event of absence.

The table below summarises the key risks and their corresponding mitigation strategies:

Table 2 Contingency plan

Risk	Likelihood	Impact	Mitigation / Contingency
Development delays due to academic workload	Medium	High	Weekly sprint milestones; buffers time reserved in final sprints
Data loss or database corruption	Low	High	Regular PostgreSQL backups; Git version control for all code
OCR accuracy issues with Nepali text	Medium	Medium	Use Tesseract with Nepali language pack; manual review option in UI
Third-party API unavailability (Translation API)	Low	Medium	Fallback to DeepL API; cache last known translation results
Security vulnerabilities / unauthorized access	Low	High	JWT authentication, RBAC enforcement, encrypted file storage, activity logs
Team member unavailability	Medium	Medium	Cross-training across modules; documented codebase and clear task assignments
Requirement changes mid-development	Medium	Medium	Agile Scrum sprints allow iterative adjustments; Product Backlog updated each sprint

2.8 Resource Requirements

SajiloDocs is developed and tested successfully based on the existence of the right hardware and software. These resources contribute to the complete project cycle such as design, frontend development, backend development, database management, OCR development, API integration, testing and documentation. Preparation of resources will minimize any inconvenience of delays that may occur unexpectedly.

2.8.1 Hardware Requirements

The hardware of this project will include personal computers systems that will be used by every team member in the process of developing the project, mobile hardware that will be used in the process of testing responsive design. A fairly capable machine is required due to the fact that the project will entail the operation of the Django backend server and a PostgreSQL database, a React frontend development server, and Tesseract OCR processing at the same time to facilitate a smooth multitasking process during development.

Table 3 Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3 / AMD Ryzen 3	Intel Core i5/i7 or AMD Ryzen 5
RAM	4 GB	8 GB or above
Storage	50 GB HDD	256 GB SSD
Internet	Broadband (5 Mbps)	Stable broadband (20 Mbps or above)
Display	1366 x 768 resolution	1920 x 1080 (Full HD)
Mobile Device	Any Android/iOS smartphone	For responsive design testing

2.8.2 Software Requirements

The software requirements of SajiloDocs include the entire development stack and all the supporting tools required throughout the project life cycle. The proposed system architecture, as reflected by the technology selections, includes a React based front-end, a Django REST Framework back-end and a PostgreSQL database served by OCR, translation and PDF generation libraries.

Table 4 Software Requirements

Category	Tool / Technology	Purpose
Operating System	Windows 10/11 or Ubuntu 22.04	Development environment
Frontend Framework	React (Vite)	Building the user interface
Backend Framework	Django (Python)	Server-side logic and API
Database	MySQL 17	Structured data storage
Code Editor	Visual Studio Code	Primary development IDE
Version Control	Git + GitHub	Code management and collaboration
OCR Library	Tesseract OCR (Python)	Text extraction from scanned documents
Translation API	Google Gemini	Nepali to English translation
PDF Generation	Python	Recreating translated documents as PDF
Authentication	Django REST Framework	Secure user login and token management
API Testing	Postman	Testing backend API endpoints
Diagram Tools	Draw.io	Creating ERD, Use Case, Flowchart diagrams

The combination of these hardware and software will offer a full and clear-cut environment of developing, testing, and demonstrating the SajiloDocs system within the given academic sche

3 Methodology

This chapter explains the used to develop the SajiloDocs system in the organized and systematic way and it also describes the how the project moved from understanding the problem to designing developing and testing the final system. A proper methodology helps to ensure that the project is completed efficiently and meets its objectives.

3.1 Considered Methodology

The development of the SajiloDocs System, several software development methodologies and were carefully evaluated to the determine the most suitable approach and the main methodologies is considered were the waterfall Model, Spiral Model and Agile Methodology.

The **Waterfall Model** is linear and sequential in nature, in which each stage, like requirement analysis, design, implementation, testing and deployment must be done before moving to the next. Although this model provides clear structure and strong documentation, it is not flexible. Any alteration in requirements at the later phases can be challenging and expensive to effect and thus it is not applicable in dynamic projects (Tutorialspoint, 2026).

Spiral Model is a method that integrates risk estimation with an iterative development. Large and high-risk projects can be effectively applied through it because it is oriented towards continuous assessment and managing risks. It is, however, comparatively very complicated and time consuming and hence not suitable when the time and resources of the project are limited like in the case of small academic projects (Art of Testing, 2021).

The **Agile Methodology**, on the contrary, emphasizes on iterative development, flexibility and constant feedback. It enables the development team to be flexible to evolving needs and provide working parts of the system in small bits. Agile encourages teamwork, expedited delivery and good quality of the product (Atlassian, 2026).

Having discussed these methodologies, Agile was chosen as the most suitable mode of handling this project (SajiloDocs) because it is flexible, efficient and capable of adapting to any changes in the project.

3.2 Chossen Methodology

The SajiloDocs is the projects adopts Agile methodology and using the scrum framework, which is used for developing the modern web applications.

Agile methodology is an iterative and the incremental approach that can divides the project in small units that's called sprints. Typically, one to two weeks for each sprint and focuses on deliver a specific feature or the module of the system, such as authentication, document management, OCR processing and the notification system. This ensures the continuous progress and the regular delivery of the functional components.

The Scrum framework is defining three key roles:

- **Product Owner:** Responsible for the managing the product backlog and the defining system requirements.
- **Scrum Master:** Facilitates the process and to ensures Agile principles are followed and remove obstacles.
- **Development Team:** Develop, designs and the tests of the system features.

The Process of begins with creation of product backlog and includes all the project requirement and features and during the sprint planning team will selects tasks from the backlog to work in upcoming sprint and forming the sprint backlog. Throughout the sprint the team will conducts daily standup meetings and discuss the progress of the project and challenges, plans and the end of sprint, a sprint review is conducted to the present the completed the work to stakeholders and gather feedback. This followed by sprint Retrospective where the team evaluates their performance and the identifies the improvements for the future sprints.

The Agile Scrum approach to ensures the continuous improvement and better collaboration and flexibility in handling changes and it also helps in the early detection of the issues and ensures the final system meets the user requirements effectively.

Overall, The Agile scrum methodology provides a structured yet adaptable framework that supports the efficient development and the successful delivery of the SajiloDocs System.

4 System Design

The suggested system also works with a simple three-layer deployment that is simple to acquire and expand. The uppermost layer is known as the Presentation Layer or front end which is the user interface everyone is familiar with such as a neat fast interface that is compatible with phones, tablets or computers (Department of Computer Science and Engineering, Michigan State University, 2025). It works with such tools as React to receive prompt updates, Angular to have smooth applications, HTML to structure them, CSS to appear, and JavaScript to have clicks and actions, and anyone can easily use it without mixups.

All the intelligent work is done in the middle Application Layer or backend. It receives user requests, enforces business rules such as calculations or checks, secures logins with authentication, receives and validates input to prevent errors or hacking, and ties all of this together with secure RESTful APIs and HTTP. It is constructed using Django, a simple Python tool, and thus development is quick, safe, and dependable.

The bottom Data Layer contains all the information in a MySQL database, which categorizes information, stores it, finds and secures information in the database in order not to lose or be mixed up. The split design allows the data to flow unimpeded between user and storage and back and enhances security through layer of protection, ease of fixing as you can only touch a single part and easy extend to more users or new features in the future(Pressman, 2021).

4.1 System Architecture

A simple, but effective 3-tier client-server architecture is used to develop the system that delineates responsibilities to improve performance and management. The user begins by the Web Application, also referred to as the frontend, when he opens his browser on any machine. This frontend enables a user to have a user-friendly interface where he or she can undertake actions such as logging in, filling in forms, browsing data, or gaining updates. Each time a user performs any action, e.g., clicking a button or typing in information, the web app makes an HTTP request through the internet to the Django Server which is the powerful backend. This request has all the required information concerning what the user wants to do. The HTTP request is sent to the Django Server that in turn takes the request and executes the relevant business logic. This involves checking the input data, enforcing task-specific rules (such as calculations, permission checks, workflow steps, etc.) and then talking to the MySQL Database, as desired. The server may access existing records, add new details safely or modify the data present in the database tables or not depending on the operation. When the backend has finished the task, it will also construct an HTTP reply with the results: e.g. updated information, error alerts, or success messages and send that to the web application promptly. Obtaining this response then, the frontend will update the screen of the user in a smooth and responsive manner, which may not be accompanied by a complete page reload, owing to the current day technologies such as JavaScript frameworks. The user live updates to the changes immediately such as updated lists, new entries or visual indications which makes the experience seem smooth and natural. The multi-layered strategy makes all things in order and has well defined roles and responsibilities of each of the layers, increases security as the sensitive operations are segregated into the backend and database and the teams are able to develop, test and maintain the system effortlessly- frontend designers do not need to communicate with the backend developers and everything can be scaled by adding more servers when necessary. (Zimmermann, 2022)

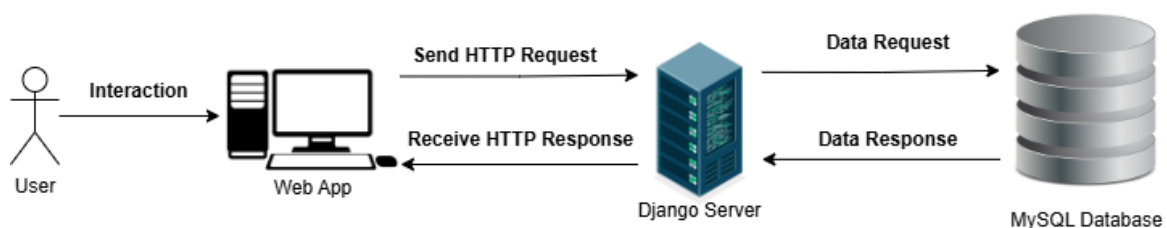


Figure 2: System Architecture of Sajilodocs

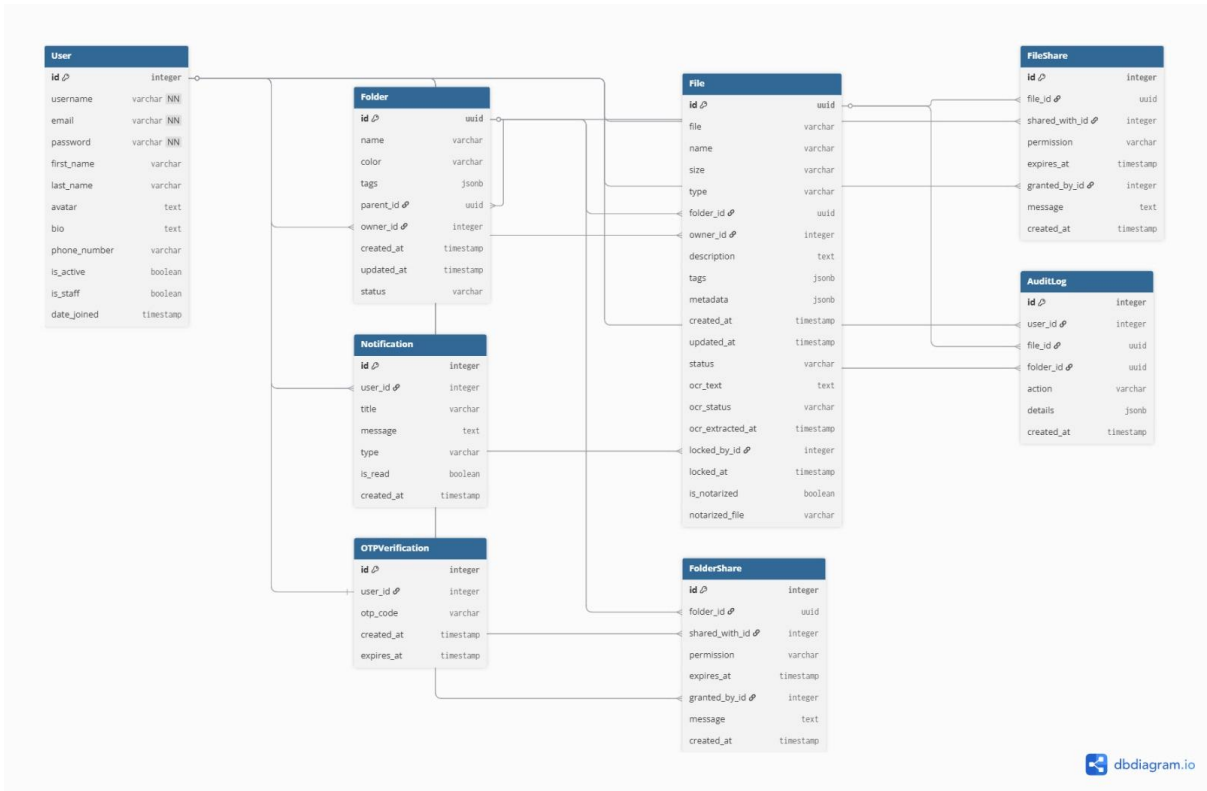
4.2 Entity Relationship Diagram(ERD)

Entity Relationship Diagram (ERD) is the full structure of data of the proposed system SajiDocs. It presents a clear identification of all the significant entities and their attributes and the association between the entities and a firm basis on which to base the design and implementation of a database. The main elements that are specified in the ERD include User, File, Folder, FileShare, FolderShare, Notification, OTPVerification and AuditLog.

One User is allowed to have several Folders and Files. The hierarchical folder structure is aided by parent-child relationship (via the parent id field) which enables the user to arrange his or her documents in a sensible and efficient tree-like way. Through the FileShare and FolderShare entities, both the Files and Folders can be safely shared with other users. Such sharing modules provide an owner of a document with the opportunity to specify certain access rights (view-only or edit), set a time frame after which the shared link is to expire, and provide additional personalized messages to the recipient.

The Notification entity will notify users of essential activities of the system e.g. successful file sharing, successful OCR text extraction, document translation results, and other system generated notifications. OTPVerification is the body that is dealing with the secure account registration as well as the process of logging in since it is managing the generation and validation of the one-time passwords sent either to the registered email address or SMS of the user. Moreover, the AuditLog entity captures all the significant activities carried out in the system such as file uploads, sharing grants, file edits, file deletion, and permit amendments, as well as information such as the user who has done t(Chen, 2025)he activity, the time, and the nature of the activity.

Such clearly defined relationships will ensure that the information is stored correctly, connected effectively and controlled with complete referential integrity in the database. The ERD does not create redundancy of data and avoids inconsistency by keeping appropriate foreign key constraints and normalization. An effective and detailed ERD is essential in ensuring an evenness of data, to provide a seamless system implementation, a well-planned enhancement in the future, and being able to maintain the SajiDocs platform in the long run.(Chen, 2025))



4.3 Use Case Diagram

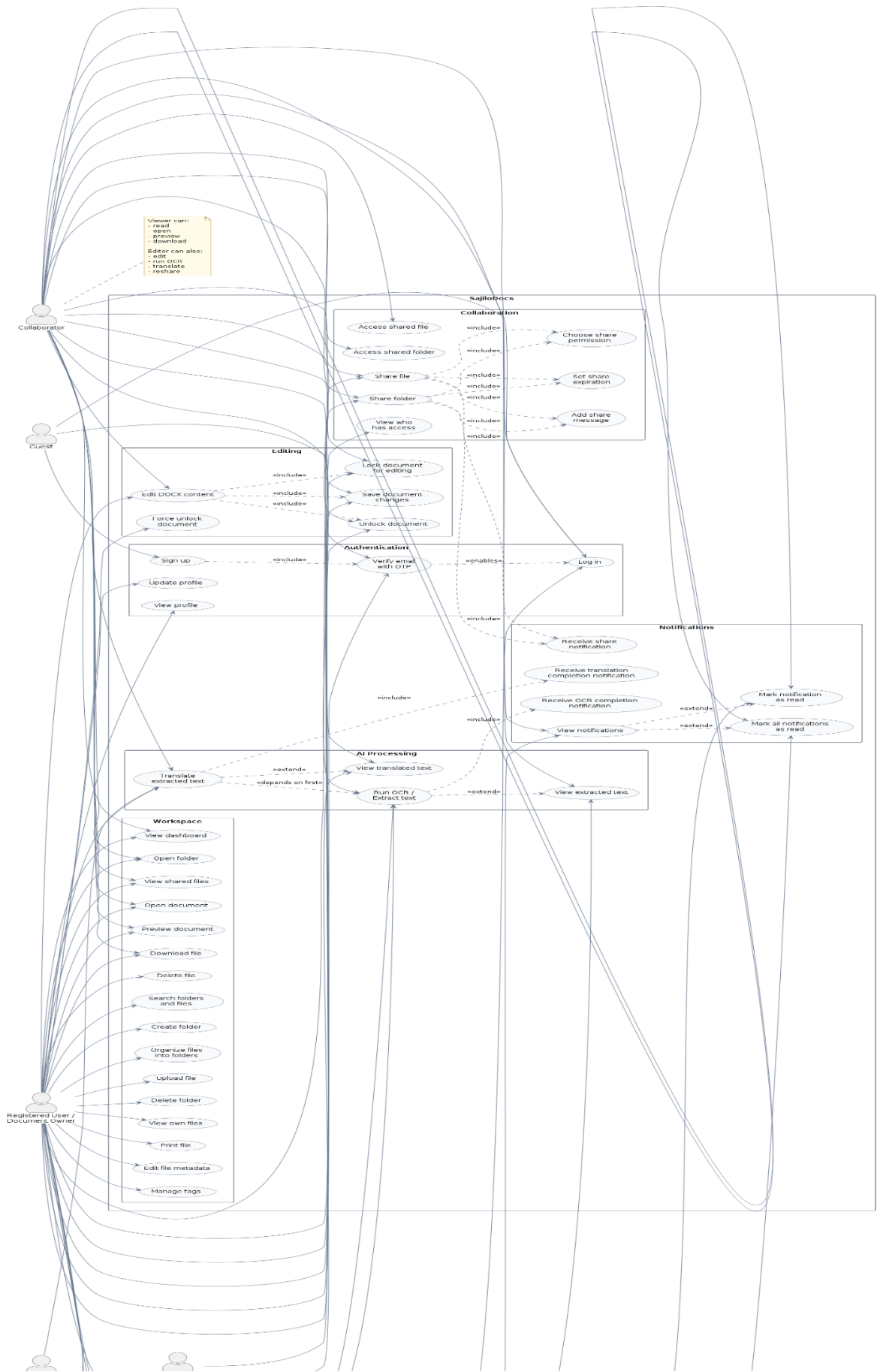
The Use Case Diagram shows the interaction of various actors with the SajiDocs system to carry out various activities and accomplish their goals. It illustrates the roles and responsibilities of each of the actors and shows the main functionalities of the platform. The main participants of the system are Registered User / Document Owner, Collaborator and Guest.

Primary users of the system are Registered Users / Document Owners. They will be able to execute a new account, log-in through OTP verification and have a full workspace. Inside the workspace, they are able to upload files, create and structure folders, view, preview, download and delete their documents. They can also edit the contents of documents, and perform advanced AI-driven services like OCR (Optical Character Recognition), to extract text and translate documents. Moreover, registered users can share files and folders with others with proper permission and expiry date, and they get real time notification of key activities such as sharing updates, OCR finally or translation output.

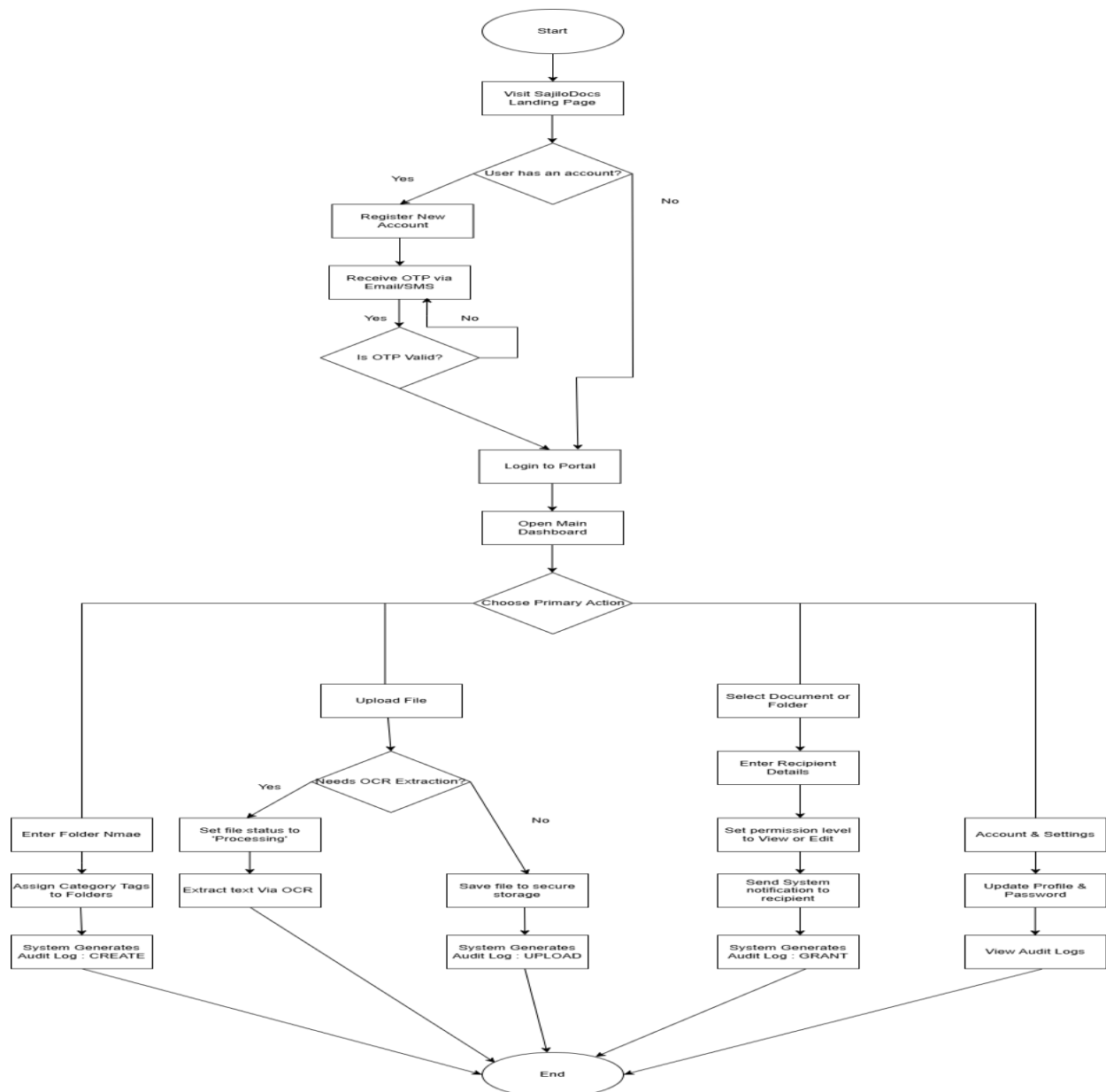
Collaborators can collaboratively access. They are able to access and read common files and folders and depending on the permissions of the actual document owner, can edit documents, do OCR or translation and other permitted actions on the shared material.

Visitors get the most limited access. They are not allowed to edit, upload, or manage any files, and only watch and download those objects that are either publicly shared or shared specifically.

This Use Case Diagram is useful in defining the various user roles, their permissions and the scope of the entire project. It assists in seeing what the system can and cannot do through the eyes of the person using it and makes sure that the key aspects of the system, such as functions and interactions, are all taken into consideration at the design stage. (Visual Paradigm Staff, 2026b)



4.4 Flow Chart



The flowchart describes the general system flow of the SajiDocs platform since the initial point of contact with the system when the user enters to the final point when all the chosen actions are performed. It begins when the user visits the opening page of SajiDocs. In case the user does not have an account, the system will generate a new account, and an OTP will be sent to the user via email or SMS depending on the option he or she chooses. Upon successful OTP entry and validation, the user may log in to the portal and get to the main dashboard. The user can choose the necessary main choice in the dashboard, i.e. uploading a file, uploading a document or a folder to share, or account and settings. As part of the processing steps, when a user uploads a file (or document/folder), the system makes a first check of whether the OCR

(Optical Character Recognition) is needed on the material uploaded. In case of the need of OCR, the system changes the status of the file to Processing, scraps off the corresponding text in the document and labels the text with category tags to enable easier search and handling of the information. Moreover, an audit log record is generated in order to trace the operation by CREATE action. But when OCR is not needed, the system bypasses OCR retrieval stages and simply stores the file to a safe storage, but still produces an UPLOAD audit trail so as to provide traceability to the action that was undertaken by the user.

Then the user, once sharing a document or folder, the user gives out information about the recipients (who will be receiving the shared item) and the permission level required, which usually is view access or edit access. After the sharing details are verified, the system notifies the recipient to inform him/her of the access that is granted. It is at this point that a GRANT audit record is created and it records that sharing permissions have been granted in a secure and traceable way.

Under the account and settings tab, the user can also change their password and update their profile or access audit logs, depending on what the platform offers under the account settings facility. Audit logs are used to record all the key tasks throughout the whole workflow, which include account creation, OTP verification, uploads, OCR processing, sharing actions, and updates to the account, among others. This is to provide the security and efficient monitoring of the user activities as well as system activities. Lastly, the process is finalized at the endpoint when the action that was chosen is done successfully, and all the information is stored safely in the system.(Visual Paradigm Staff, 2026a)

5 Testing

Testing is a process where we verify if our code behaves as intended, ensuring each part of a program produces the expected results and continues to function correctly when changes are made. Rather than assuming that the code works, testing confirms it through repeatable, automated tests. It plays a central role in modern software development by catching bugs early, preventing regressions, and building confidence in new features (Mimo, 2026).

The testing was done to ensure that the developed SajiloDocs system is working as per the specified requirements and it is able to handle interactions with the users as expected. The user authentication, document upload, OCR processing, translation, sharing controls and expiry notification modules are some of the main modules tested within normal and exceptional conditions to determine whether errors are embedded in the modules being tested, whether the system output is validated and to enhance the reliability of the system.

The testing strategy was oriented to functional testing, in which every functionality was put to test with valid and invalid inputs to check appropriate system behavior. The system was put to test in a web based system with modern browsers like Chrome with the backend running on Django and MySQL database.

The intention of testing was to verify that all the functionalities of the system are functioning properly, that there is a safety of sensitive document data, and that the platform is stable and demonstrable and ready to engage in actual use.

5.1 Test Case 1

Table 5: Test 1- User Registration

Field	Description
Test Case ID	TC-01
Module	User Registration
Input	Valid name, email, and password
Expected Result	Account is created and redirected to the dashboard
Actual Result	Account created successfully
Status	pass

Test Case 1 was prepared to check the functionality of the feature of user registration when a user inputs the correct information and anticipates the creation of the account. Its anticipated outcome is that the system will save the information belonging to the user as safely and send the user back to the dashboard. The result obtained was in line with the behavior expected, which implies success.

5.2 Test Case 2

Table 6: Test Case 2- User Login

Field	Description
Test Case ID	TC-02
Module	User Login
Input	Incorrect email/password
Expected Result	Error message displayed, incorrect user credentials
Actual Result	Error message shown correctly
Status	pass

Test Case 2 was designed to test a system of incorrect log-in attempts. The user gets a username and a password that is incorrect. The system is expected to not allow access and give a relevant error message. The result that was gained was anticipated.

5.3 Test Case 3

Table 7: Test Case 3-Document Upload

Field	Description
Test Case ID	TC-03
Module	Document Upload
Input	Valid file (PDF/JPG) with metadata
Expected Result	File uploaded and displayed in the dashboard
Actual Result	File uploaded successfully
Status	pass

The goal of Test Case 3 was to test the uploading of documents. A useful document file (PDF/JPG) with metadata will be posted by the user. It is anticipated that the file would save and would appear in the user dashboard. The real outcome was satisfactory.

5.4 Test Case 4

Table 8: Test Case 4-Text OCR Extraction

Field	Description
Test Case ID	TC-04
Module	OCR Processing
Input	Scanned document image
Expected Result	Text extracted and displayed correctly
Actual Result	OCR output displayed successfully
Status	pass

Test Case 4 was developed keeping in mind to test the OCR of the system. The user uploads an image of a document, which has been scanned, and the system will be able to extract the text in OCR. It is assumed that the system will offer the appropriate functionality of the task of extracting the readable text and detecting the ultimate fields which include document number and dates. The result observed was as expected.

5.5 Test Case 5

Table 9: Test Case 5-Document Translation and PDF Generation

Field	Description
Test Case ID	TC-05
Module	Translation & PDF Generation
Input	Extracted Nepali text
Expected Result	Translated text displayed and PDF generated
Actual Result	Translation and PDF generated successfully
Status	pass

Test Case 5 was to test the translation feature. The system converts Nepali writing extracted into English and the output is a formatted PDF. One would expect that the content translated can be downloaded and is correct. This was the real outcome as expected.

5.6 Test Case 6

Table 10: Test Case 6-Expiry Notification System

Field	Description
Test Case ID	TC-06
Module	Expiry Notification
Input	Document with expiry date set
Expected Result	Notification sent before expiry
Actual Result	Notification triggered successfully
Status	pass

Test Case 6 has been developed to test the expiry alert feature. A document that has an expiry date which is near is stored in the system. This system is expected to alert the user about the expiry date.

6 Group Contribution Table

Table 11: Group Contribution Table

Member Name	LMUID	Major Contributions	Key Responsibilities	Contribution %
Samir Nepal <i>Backend Lead</i>	24046042	- Requirement Gathering (Lead) - Product Backlog Creation - Methodology Selection (Scrum) - Authentication API — Sprint 1	- Core backend API development across all sprints - JWT token management & authentication - OCR and Translation API integration - Role-Based Access Control (RBAC)	25%

		• Document CRUD API — Sprint 2 • Permissions API — Sprint 3 • OCR Engine Integration (Tesseract) — Sprint 4 • Translation API — Sprint 5 • Search API — Sprint 6 • Notification API — Sprint 7 • RBAC Hardening — Sprint 8 • Final Deployment Setup (Release)	• Full-text search implementation • Production deployment	
Susant Kulung <i>Frontend Lead</i>	24046121	• UI/UX Design & Wireframes • Product Backlog Creation • Login & Register UI — Sprint 1 • Document Upload UI — Sprint 2	• All frontend screens and UI components • Initial wireframes (dashboard, OCR, translation screens) • Responsive design using Tailwind CSS • File encryption setup	25%

		• Share Link Generation UI — Sprint 3 • OCR Result Screen UI — Sprint 4 • Translation Review UI — Sprint 5 • Search & Filter UI — Sprint 6 • Reminder Creation/Update UI — Sprint 7 • Encryption Setup — Sprint 8 • Frontend Deployment & Final Fixes • User Guide Documentation (Release)	• API documentation and user guide	
Manish Bhattarai <i>Database Lead</i>	24045919	• Technical Feasibility Support (Initiation) • JWT Refresh Token Integration — Sprint 1	• Full database schema design and implementation • All SQL table creation across all sprints • PDF generation using ReportLab	25%

		• SQL Documents Table — Sprint 2 • SQL Permissions Table — Sprint 3 • SQL OCR Data Table + Auto-field Detection — Sprint 4 • SQL Translation Table + PDF (ReportLab) — Sprint 5 • SQL Search Index Table — Sprint 6 • SQL Reminder Table — Sprint 7 • SQL Activity Log Table — Sprint 8 • Backend/OCR Final Fixes (Release)	• JWT backend token logic • OCR auto-field detection logic • Activity logging system	
Ganga Bajgain	24045880	• Project Charter Preparation (Initiation)	• Test case execution across all sprints	25%

<i>QA & Planning</i>		• UI/UX Design Support • Authorization Testing — Sprint 1 • Integration Testing — Sprint 2 • Access Log Tracking — Sprint 3 • OCR Accuracy Testing — Sprint 4 • Translated PDF Testing — Sprint 5 • Search Results Testing — Sprint 6 • Email & In-app Alert Testing — Sprint 7 • Security Testing — Sprint 8 • User Guide & API Documentation (Release)	• Project charter and scope definition • Access log and security verification • OCR and translation quality assurance • Final documentation and user guide	
--------------------------	--	---	---	--

The project was done as teamwork with every member playing a role in various stages of the project such as initiation, all the eight development sprints and release phase. Despite the

distribution of the responsibilities based on the personal strengths (Samir worked on backend APIs, Susant worked on frontend UI, Manish worked on the entire database layer and Ganga worked on the QA and project planning), the members of the team actively engaged in daily stand-ups, sprint reviews and retrospectives. The percentage of contributions represented are a viable and balanced level of involvement of each member towards the successful completion of the SajiloDocs project.

7 Conclusion

This project was done to address the issue of how to organize and manage important personal and legal documents in a secure, organized and efficient manner. Even in the digitalized world nowadays, individuals continue to encounter problems concerning storage, retrieval, and transfer of documents including citizenship certificates, academic reports, and legal documents. Available solutions such as cloud storage services only offer general storage and not intelligent features such as document categorization, text extraction using OCR, translation, and tracking of document expiry. These constraints usually result to lack of organization, wasting of time, and security fears.

To overcome these complications, the project SajiloDocs was created and became a smart digital document management system. The system has a centralized system whereby users easily upload, tag and manage their documents. It has components like the secure user authentication, categorization of documents, OCR text extraction, automatic translation of Nepali to English, expiry notifications, and role-based sharing of documents. These characteristics render the system not only useful but also applicable in a real-life context.

The system was developed in systematic approach in accordance with Scrum framework. The process enabled the project to be created over a series of iterations, which guaranteed the construction of the project under constant enhancement and the correct management of the requirements. All modules such as authentication, document processing, OCR, translation, notification and security were done in steps, with a system to test its reliability and performance.

The last system is able to achieve its prime goals. It increases the accessibility of documents, minimises manual efforts, increases security and offers user-friendly experience. The user is now able to store and retrieve documents in a short time, get reminders in good time and share documents with controlled access. It has added great value with the incorporation of OCR and translator features, particularly when the Nepali users need to convert and comprehend the documents.

Besides the technical deliverable, other key skills that were gained through this project include problem analysis, system design, teamwork, and full-stack development. The experience of working with such technologies as React, Django, and MySQL gave the practice in creating a full-fledged web-based system. In general, the project shows that a properly designed digital

solution may be used effectively to address the real-world issues concerning document management.

7.1 Legal Issues

The creation and application of the SajiloDocs system has a number of legal implications given the fact that it deals with sensitive personal and legal files. Data protection and privacy is one of the key legal issues. The user records which occupy the system include identification papers and certificates and must be in a secure and legal way. Care must be taken to make sure that data on users is not accessed, shared or misused without the proper authorization.

The other significant question of law is software licensing compliance. The tools, libraries, and APIs utilized in the course of the development include OCR engines and translation services. These tools should be duly licensed and they should adhere to the terms of usage to prevent any law contravention. The third-party services that are to be embedded in the system should be utilized within the legal agreements.

There is also the copyright and intellectual property rights. Documents should not be duplicated or misused by unauthorized persons in the system. The user must not post the documents that he/she does not own or that he/she is not permitted to use. In the same way, all the external materials like pictures, code libraries or references to the project should be properly attributed to prevent the use of plagiarism or copyrighted materials.

Furthermore, in case the system is scaled in the future to be deployed in reality, it has to be in accordance with the national and international legislation on data protection. This involves obtaining user permission, safe storing of information, and management of sensitive data. To make the system trustworthy and legally tolerable, there is a need to address these legal issues.

7.2 Social Issues

SajiloDocs system has several good social impacts since the system will aim at improving the way people handle important documents. The first strength is increased accessibility. Their documents can be accessed at any time and place, that is, useful especially in cases of emergencies. This eliminates the need to rely on hard copies and it also helps to avoid loss or damage.

Electronic organization and efficiency is also promoted by the system. It enables the user to organize their documents better, remind them as well as finding the corresponding documents through organization of their storage, reminder-wise and searching. It may prove particularly

useful when dealing with students, employees, and such small businesses as operate with multiple documents on a regular basis.

Yet even there are some social issues. The internet and digital devices are not necessarily as available to everyone. Some of such systems can prove difficult to work with individuals who occupy rural locations, or less technologically skilled. This causes a digital divide where other users benefit more than others.

To manage this issue, the system should be simplified and made easy to operate to ensure that it is easily operated by the non-technical users. In the future, it can be made more accessible with the design of mobile devices and low-bandwidth situations.

Overall, the system has the potential to have a positive impact on society as it helps to make it more efficient, reduces the number of papers, encourages the digital revolution, and, simultaneously, enlightens the audience about the necessity to offer equal opportunities in the realm of technology.

7.3 Ethical Issues

The reason behind this is that ethical considerations are highly essential in the development of a system like SajiloDocs which require taking of sensitive personal information. One of the most important ethical obligations is to make sure that the privacy of the users is addressed. Individuals post their prized documents on the system and in this connection, one should make sure that this information is not prone to unauthorized use and misuse.

The other ethical area is transparency. The system should clearly provide the storage, processing and use of users data. They must know about the control of their data (they are to be able to control them, they are to be able to delete their information, etc.).

Data security is closely correlated with ethics. Having good security measures such as encryption, authentication and access control measures should be implemented to protect user data. The inability to obtain the security may have harsh consequences, including identity abuse or data theft.

Another ethical issue is fairness and good use of technology. This system is not supposed to favor any group of the users and it must be designed in a way that it can be inclusive. In addition, some features, like OCR and translation, should provide the correct results since in case it gives false results, it can lead to misunderstanding or misuse of the documents.

Finally, developers are to ensure that the system is used in a legitimate way only. The measures to be taken to deter abuse should include posting of forged or unapproved documents. The system requires ethical standards to be supported which will make the system reliable, trustworthy and useful to the users.

7.4 Future Work

Despite the fact that the SajiloDocs system is effective in fulfilling the primary goals, one can outline multiple areas of improvement and enhancement in the future.

The development of a mobile application is one of the potential improvements. This would enable users to manage and access their documents more easily using smart phones. Mobile integration would allow greater accessibility and make the system more workable in daily use.

The other feature to be developed in future is advanced security features. This may involve the two-factor authentication, biometrics loading, and the enhancement of the encryption procedures in order to further secure user information. Security will be improved to raise the level of trust and reliability of the system used by the users.

It is also possible to enhance the system by incorporating cloud storage services to enhance its scalability and backup facilities. This would make sure that the data of the users is safely stored and can be retrieved in case of system failure.

The OCR and document classification features can be complemented with the artificial intelligence and machine learning. This would enhance the precision of text extraction, automatic tagging and intelligent search.

Moreover, the system can be supported with the help of multiple languages to enable more people to use it. It should be expanded in terms of translation into other languages other than Nepali and English to enhance its application in other areas.

Lastly, the system can be incorporated to interface with government or organization systems to verify and validate documents. This would render it stronger and applicable in the real world.

All in all, these future enhancements are capable of improving the functionality, usability and impact of the system to make it more advanced and be applicable.

8 References

- Chen, P. P.-S. ; S. C. (2025). *Encyclopedia of Database Systems Institution: Elsevier B.V.* 45(3).
- Dropbox. (2026, March 25). *Dropbox: Cloud Storage and File Sharing*. Dropbox.Com.
- Hanna, K. T. (2026, March 25). *What is Google Drive?* Techtarget.Com.
- Microsoft. (2026, March 25). *Microsoft OneDrive: Online Cloud Storage*. Microsoft.Com.
- Pressman, R. S. , & M. B. R. (2021). System Design. In *Software Engineering: A Practitioner's Approach (9th ed.): N/A* (N/A, pp. 123–150). McGraw-Hill Education.
- Visual Paradigm Staff. (2026a). *Visual Paradigm Tutorials Series*. Visual Paradigm International Ltd.
- Visual Paradigm Staff. (2026b). *UML Unified Modeling Language Guide Series*.
- Wadhwa, K. (2024). *The Role of Gantt Chart in the Project Management*. Laurea University of Applied Sciences.
- Zimmermann, A. S. K. S. K. J. D. (2022). A Systematic Literature Review and Synthesis on Architecting Software Systems-of-Systems. In S. (Conference C. Chung (Ed.), *IEEE SOSE 2022 Conference Proceedings* (pp. 23–33).
- Art of Testing. (2021) *Spiral Model - Definition, Phases, Advantages and Disadvantages* [Online]. Available from: <https://artoftesting.com/spiral-model> [Accessed 25 March 2026].
- Atlassian. (2026) *What is Agile?* [Online]. Available from: <https://www.atlassian.com/agile> [Accessed 25 March 2026].
- Department of Computer Science and Engineering, Michigan State University. (2025) *09a-SystemDesign-2025 Notes* [Online]. Available at: <https://www.cse.msu.edu/~cse870/Lectures/Notes/09a-SystemDesign-2025-notes.pdf> [Accessed 21 March 2026].
- LaPrad, L. (2026) *What Is a Gantt Chart? A Complete Guide to Project Timelines* [Online]. Available from: <https://www.teamgantt.com/what-is-a-gantt-chart> [Accessed 26 March 2026].
- Mimo. (2026) *Testing: Definition, Purpose, and Examples* [Online]. Available from: <https://mimo.org/glossary/programming-concepts/testing> [Accessed 23 March 2026].

Tutorialspoint. (2026) *SDLC - Waterfall Model* [Online]. Available from: https://www.tutorialspoint.com/sdlc/sdlc_waterfall_model.htm [Accessed 25 March 2026].

9 Developed Project

9.1 Screenshots

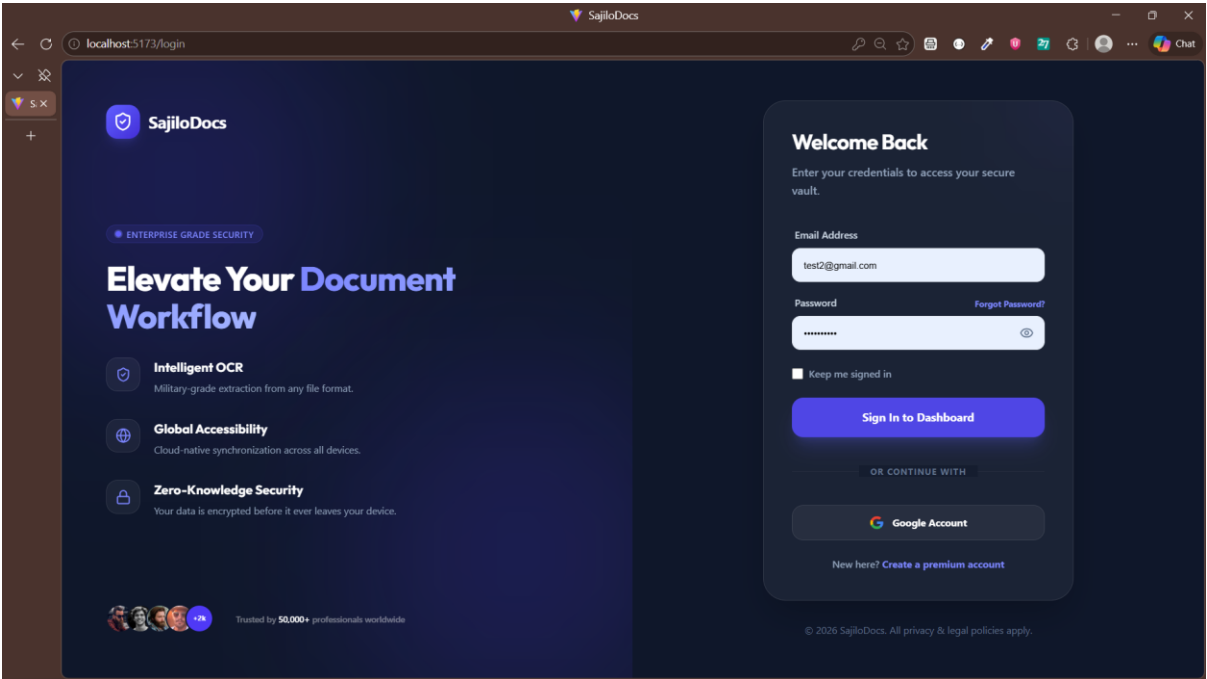


Figure 3: Login Page

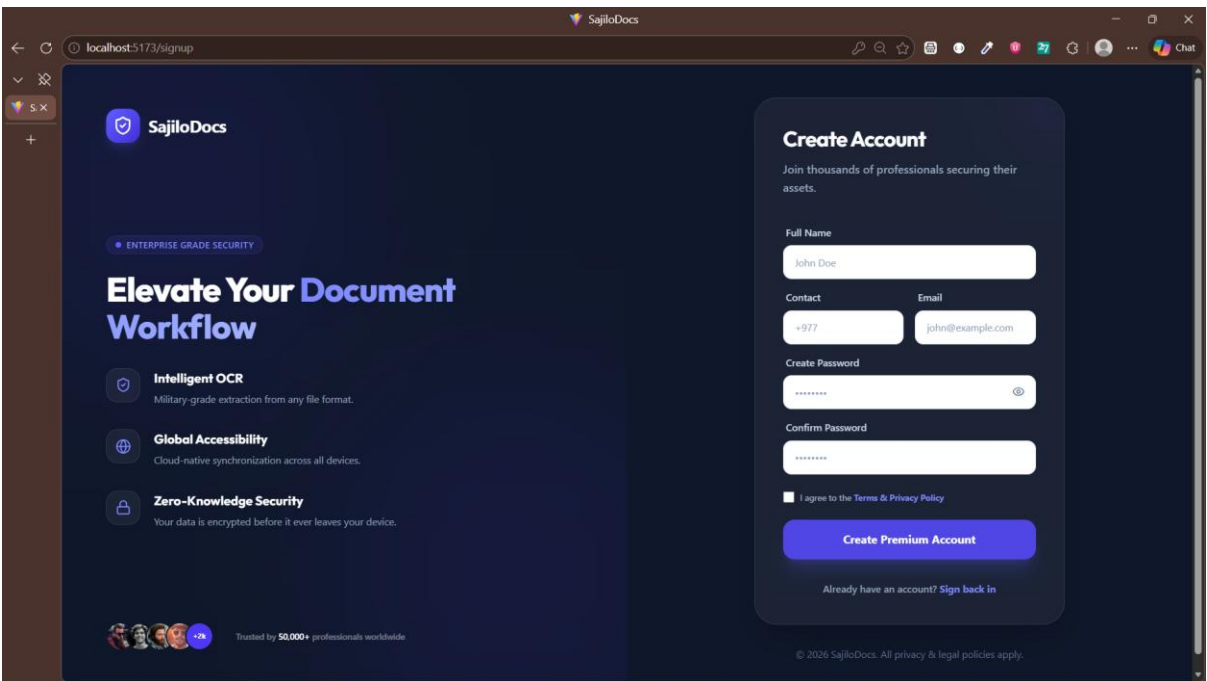


Figure 4: Sign Up Page

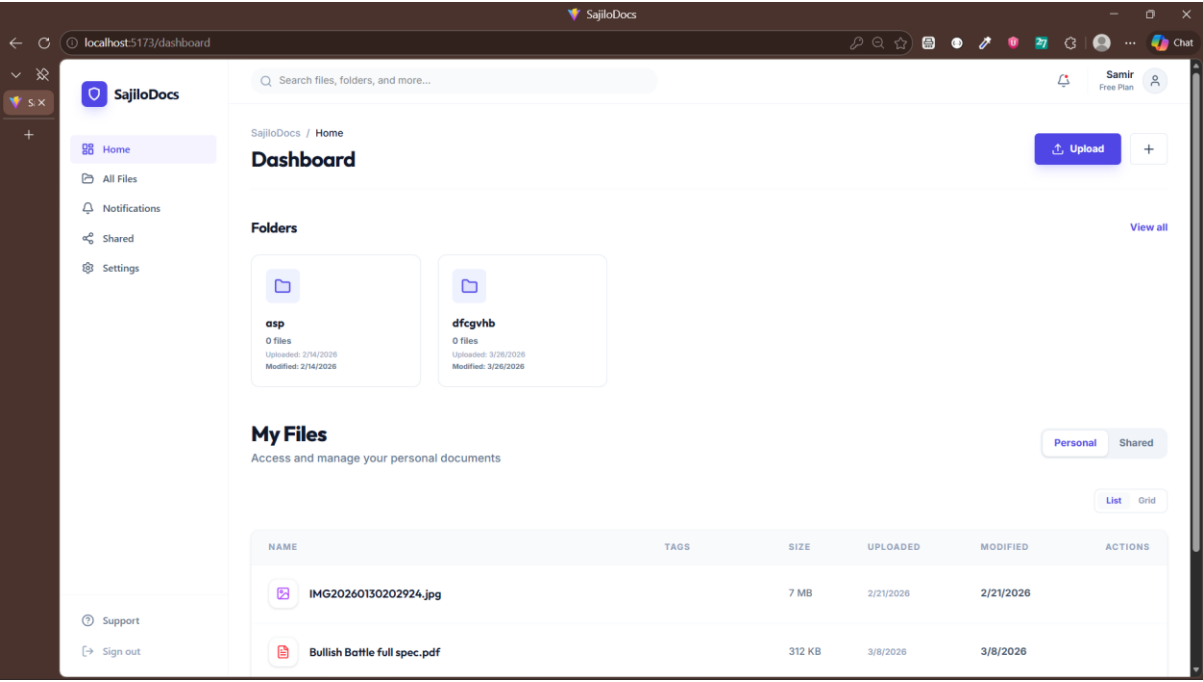


Figure 5: Dashboard Page

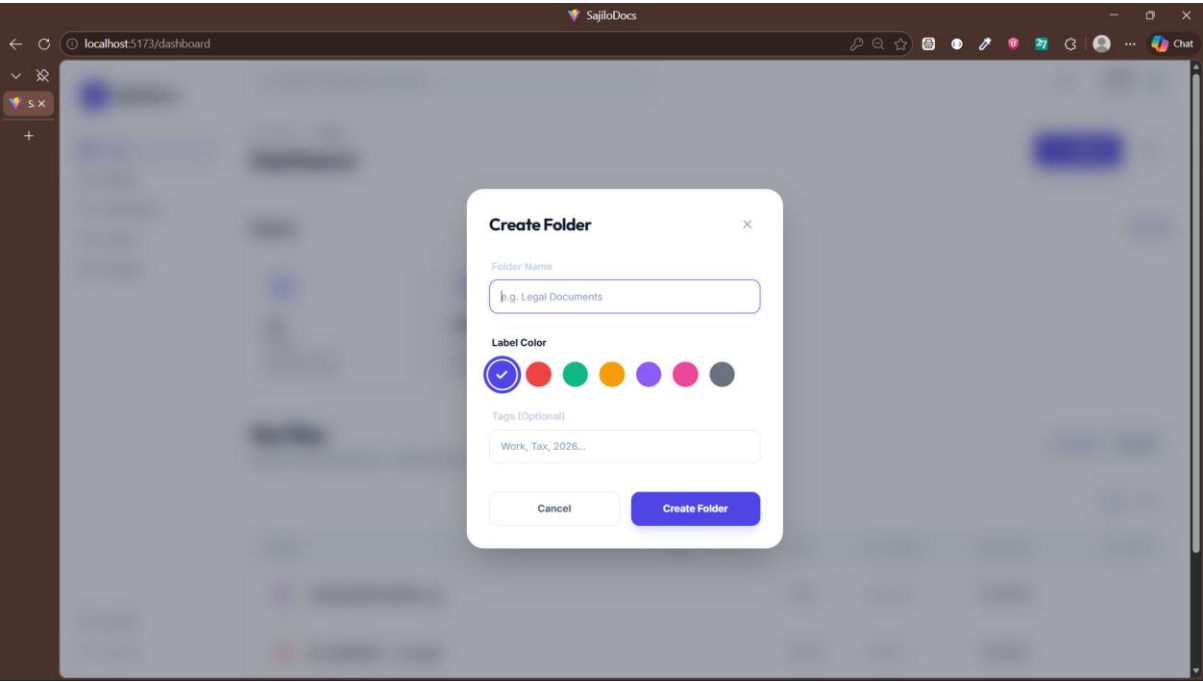


Figure 6: Create folder

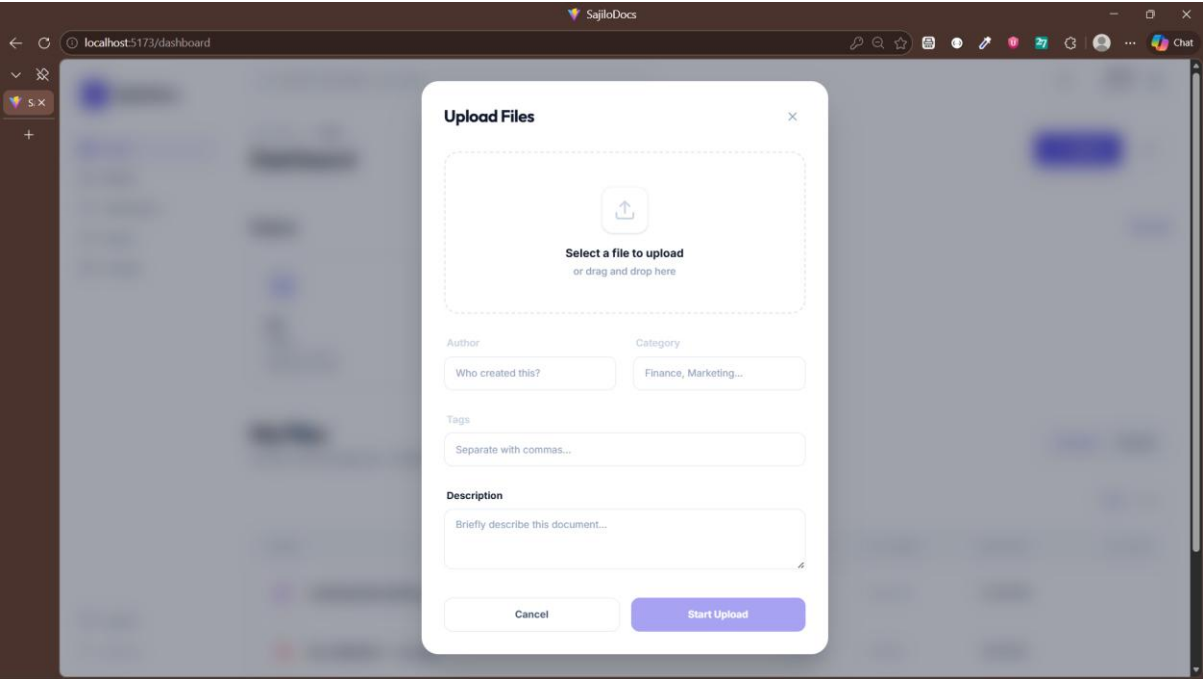


Figure 7: Upload File

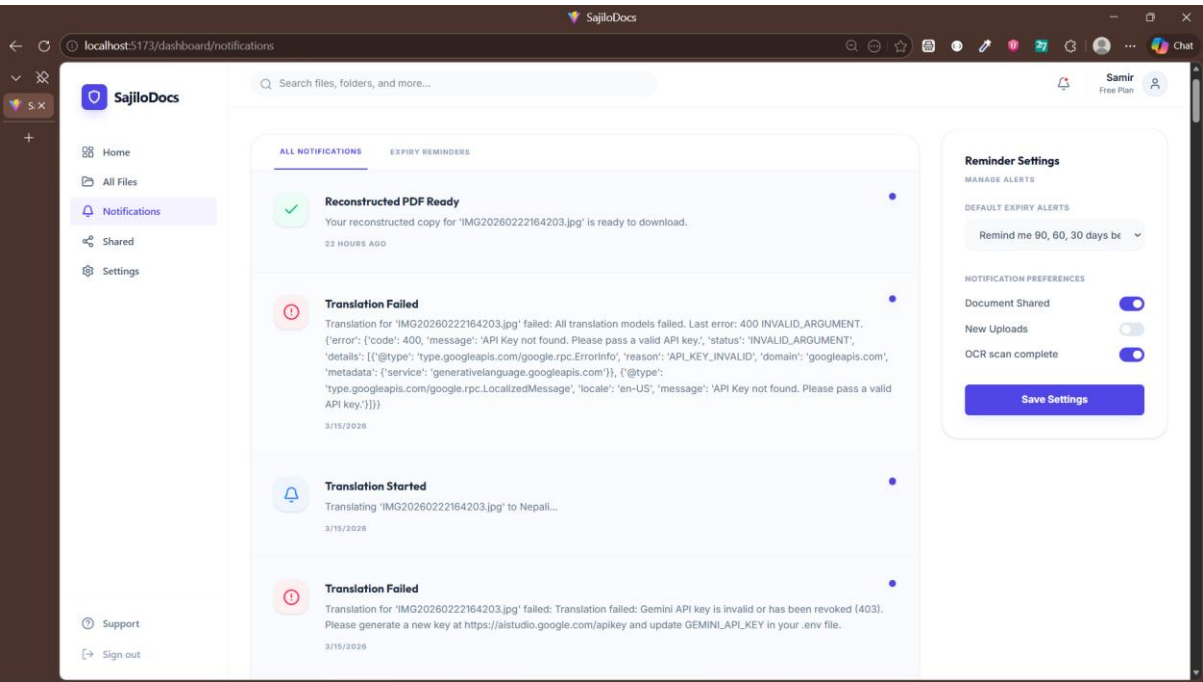


Figure 8: Notifications Page

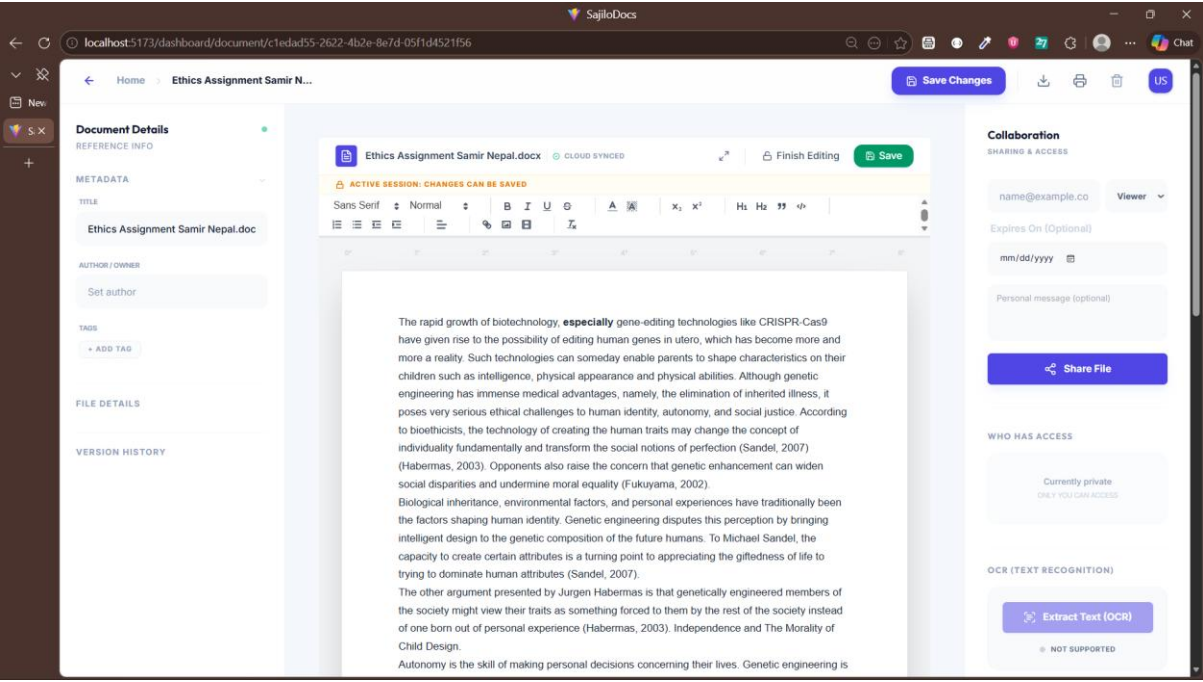


Figure 9: Docx Edit Page

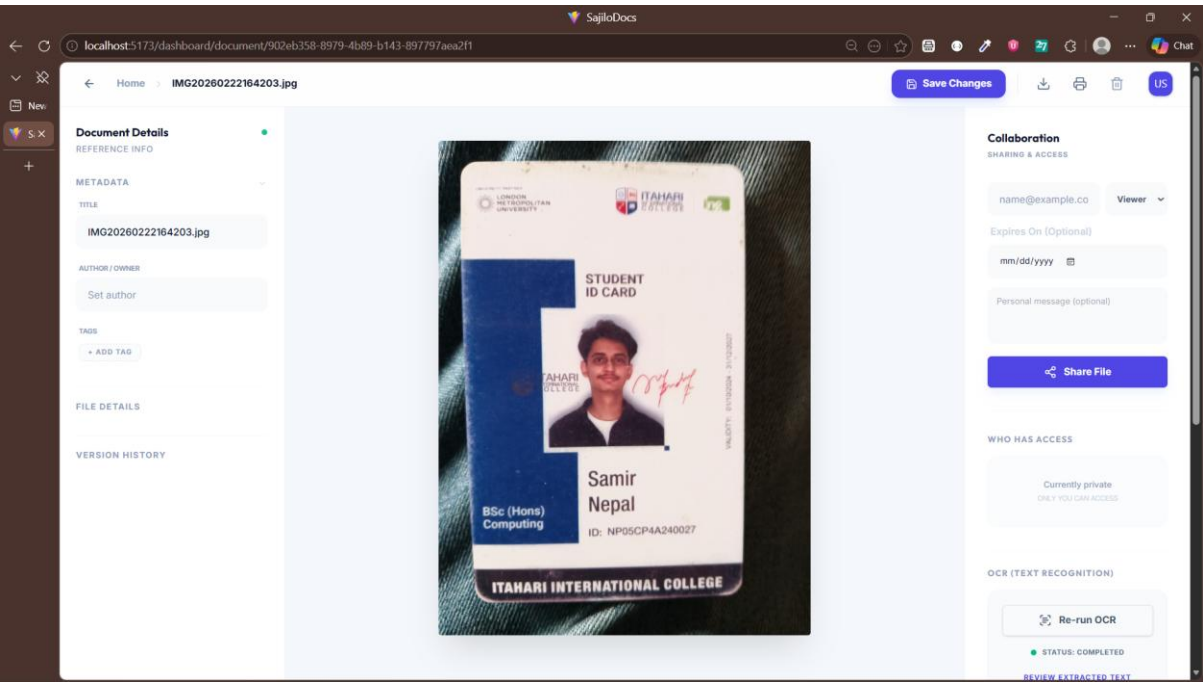


Figure 10: Image View

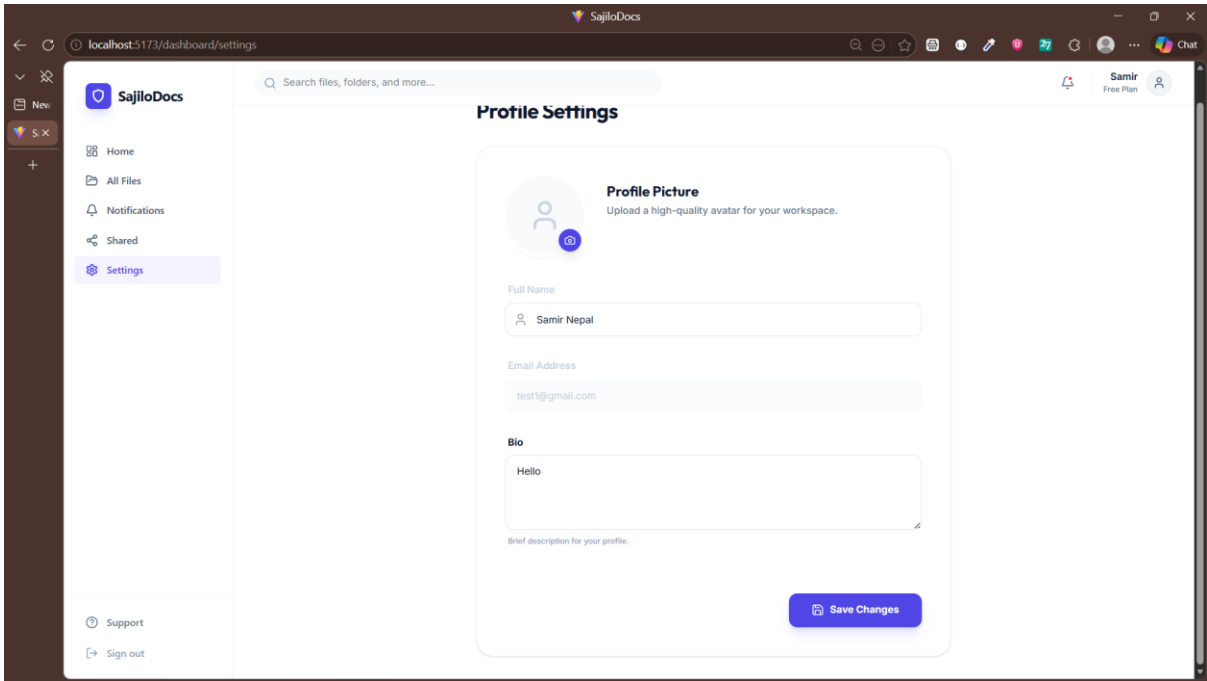


Figure 11: Edit Profile Page

9.2 UI Designs

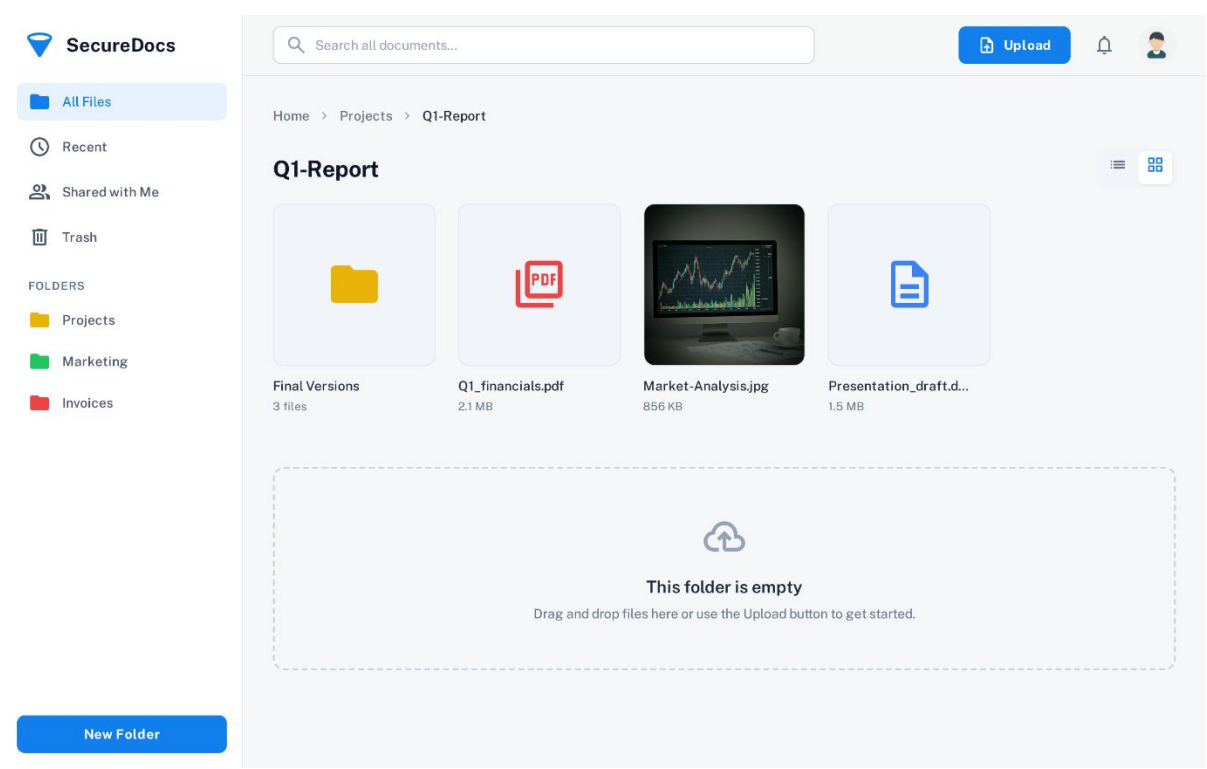


Figure 12: Dashboard UI

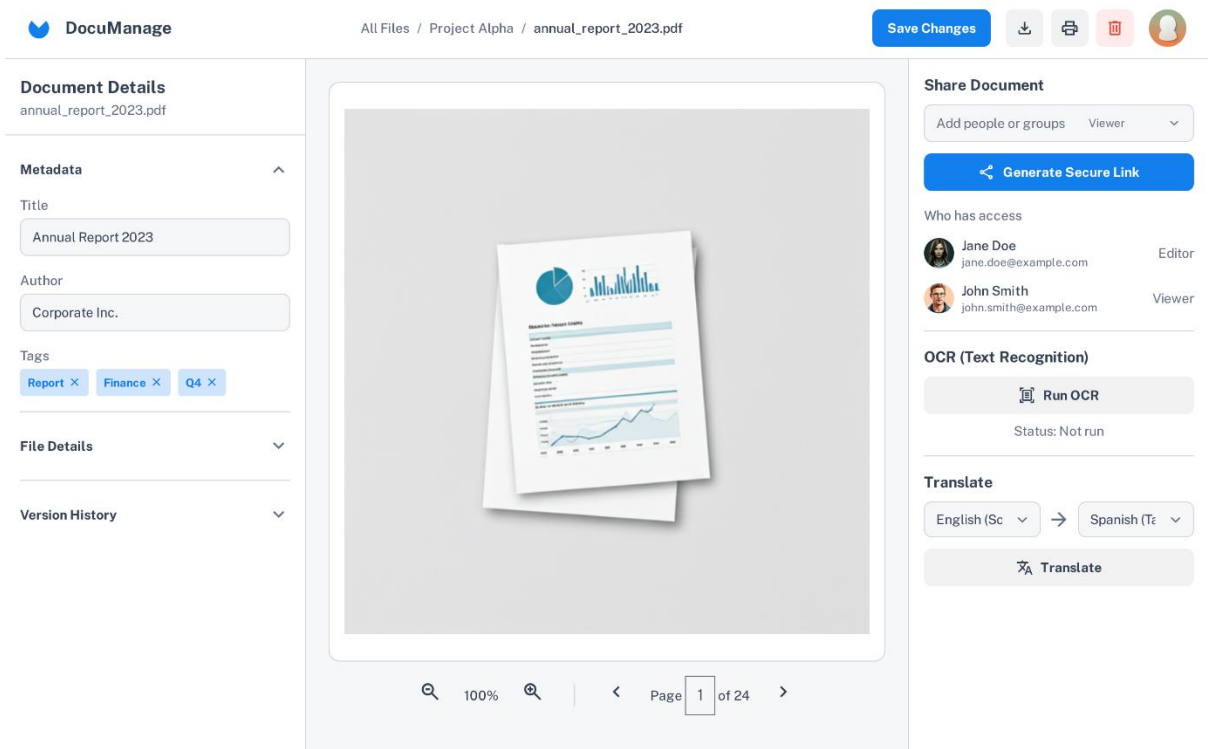


Figure 13: Document View UI

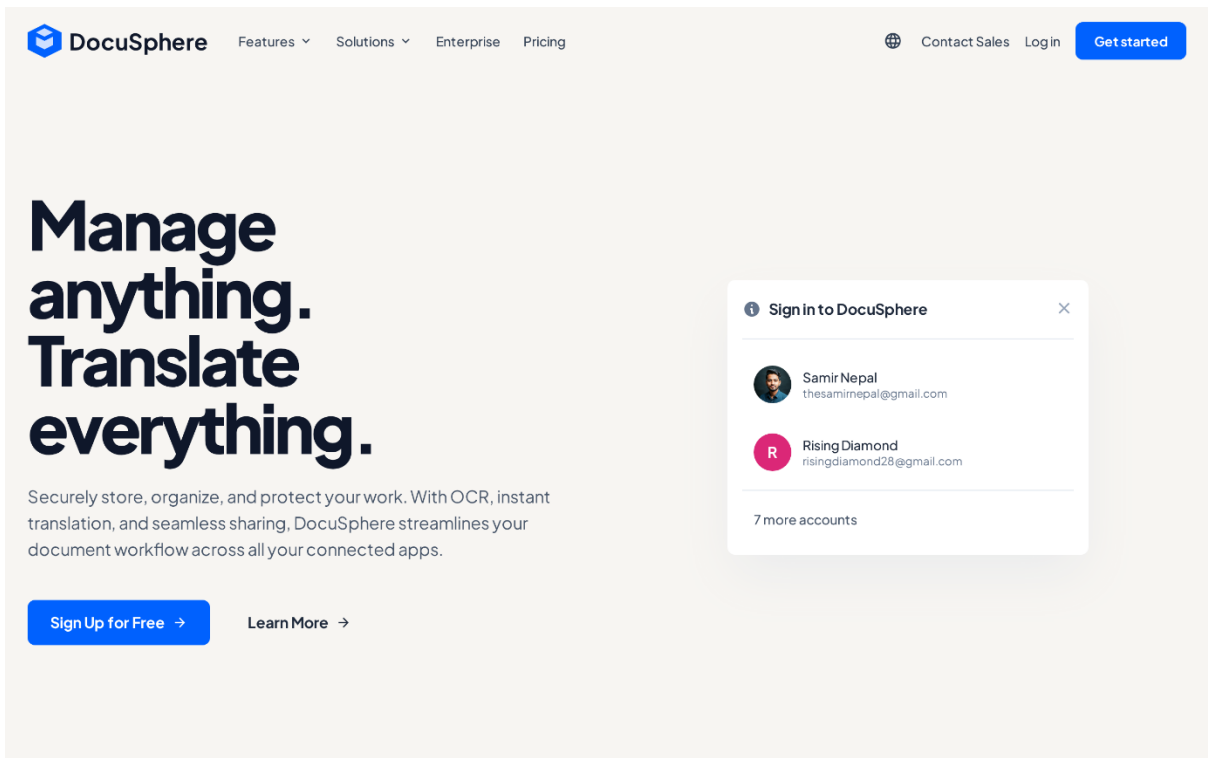


Figure 14: Home Page UI

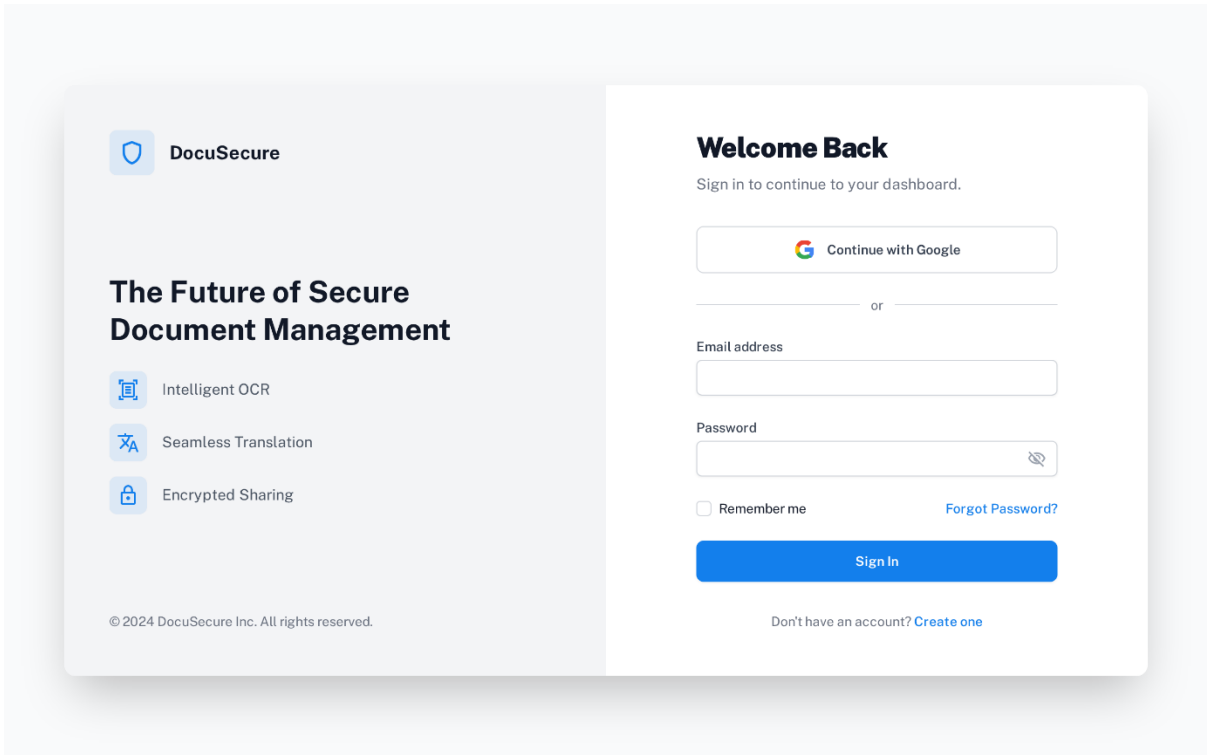


Figure 15: Login UI

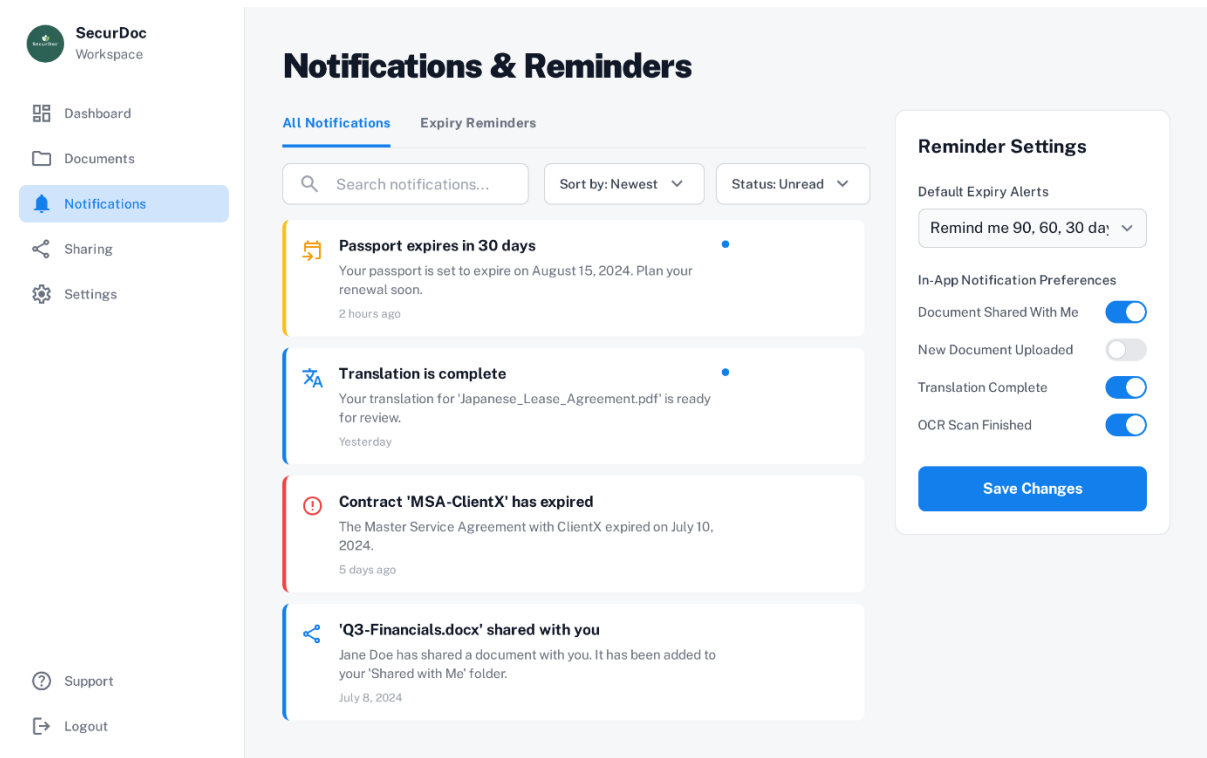


Figure 16: Notifications UI

 DocuSecure

Profile

Notifications

Preferences

Profile

Update your photo and personal details.



John Doe

john.doe@email.com

Cancel

Save Changes

Full Name

John Doe

Email Address

john.doe@email.com

Phone Number

e.g., +1 234 567 890

Account Security

Manage your password and two-factor authentication.

Change Password

Enable Two-Factor Authentication

Log Out

Figure 17: Edit Profile UI

Sajilodocs

52

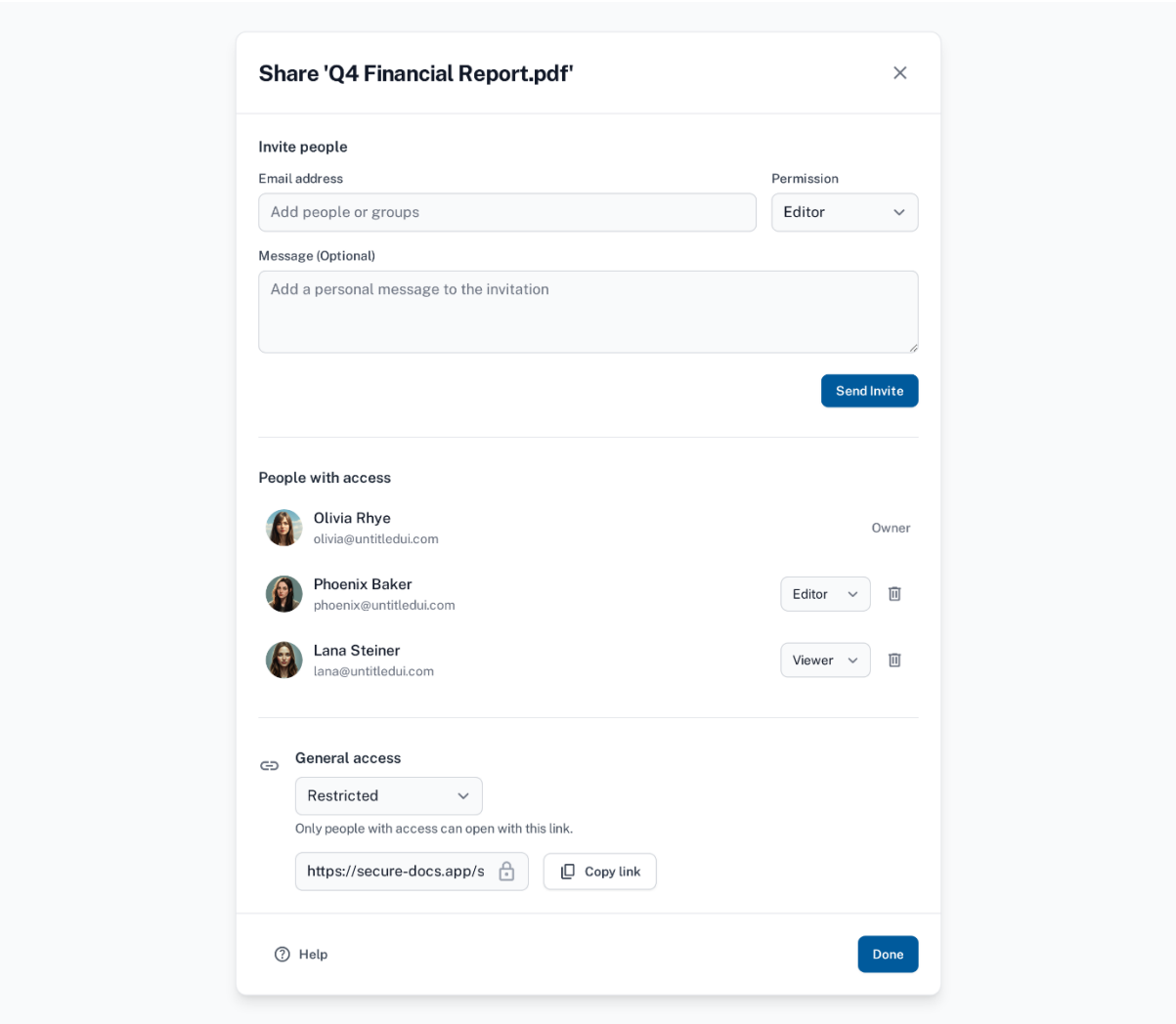


Figure 18: Give Access UI